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Variable power functional dilution adjustment of spot urine

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Spot-urinary biomarkers are crucial in medical, epidemiological, and environmental studies, but their variability due to hydration levels requires precise dilution adjustments. Traditional methods, like conventional creatinine correction (CCRC), are limited in compensating for variations in urine concentration, causing substantial inconsistencies, particularly at the extremes of the diuresis spectrum. While restricting the creatinine (CRN) range to 0.3–3 g/L is recommended to ensure result stability, this approach excludes a substantial proportion of samples and permits notable fluctuations within the accepted range. This study introduces a novel variable power functional creatinine correction method (V-PFCRC) to normalize analytes to 1 g/L CRN by utilizing uncorrected analyte levels and two analyte-specific coefficients, c and d. Based on extensive urinary total weight arsenic data (*n* = 5,553), the mathematical derivation of these coefficients is detailed in this paper and forms the foundation of the corrective V-PFCRC formulas. The generalizability of V-PFCRC was evaluated using large spot-urinary datasets for four additional metals and an extensive dataset of urinary iodine levels (*n* > 58,000) and blood iodine. Validation against conventional methods—assessing vital statistical data, residual CRN bias, and correlations with concurrently detected blood levels of total arsenic and iodine— demonstrated the superior performance of V-PFCRC in reducing residual CRN bias and enhancing blood-urine correlations. The V-PFCRC method effectively addresses nonlinear hydration bias and the exposure-dependent variability of this bias, providing a more accurate representation of exposure and supply levels. The adaptability and efficiency of V-PFCRC suggest its broad applicability across various scientific disciplines, potentially transforming the precision and reliability of urinary biomarkers.

Keywords Nonlinear dilution adjustment, Creatinine correction, Spot urine, Biomarkers, Arsenic, Iodine, Exposure studies.

Abbreviations

As	Arsenic
CCRC	Conventional Creatinine Correction (by calculating the ratio Analyte/CRN)
CRN	Urinary Creatinine Concentration (g/L)
DMA	Dimethylarsinic acid
ICP-MS	Inductively Coupled Plasma Mass Spectrometry
IP	Intersection Point, Limit of Valid Dilution Adjustment Range of V-PFCRC
IQR	Interquartile Range
MMA	Monomethylarsonic acid V-PFCRCVariable Power-Functional Creatinine Correction (using a variable exponent b)
SAME	S-Adenosyl-Methionine
SDAE	Systemic Dilution Adjustment Error
S-PFCRC	Simple Power-functional Creatinine Correction (using a fixed exponent b)
uAs_C	Conventionally Creatinine-Corrected Urinary arsenic (µg/g CRN)
uAs_N	Power-functionally adjusted urinary arsenic, normalized to µg/1 g CRN
uAs_{UC}	Uncorrected urinary arsenic (µg/L)
UC	Uncorrected Urine Results (µg/L)
UFR	Urinary Flow Rate, Urine Excretion/Time (ml/min)

Urinary dilution correction in biomonitoring

Urinary biomarkers are widely used to assess human exposure to environmental toxins, carcinogens, and pharmaceuticals across various clinical, metabolic, and forensic contexts. An optimal biomarker should precisely reflect exposure levels, indicate potential toxicity, and maintain minimal intra-individual variability under

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consistent exposure conditions. High inter-individual variability should primarily result from differences in exposure, with minimal influence from confounding factors[1]. Compared to blood, urine is more accessible and can reflect either accumulated body burden (e.g., cadmium) or short-term exposure (e.g., arsenic), depending on the toxicokinetics of the substance[2]. However, despite their extensive use, the validity of urinary biomarkers in assessing toxin load, exposure duration, and potential toxicity remains a critical question[2–5]. Toxic metals and metalloids, such as arsenic, lead, mercury, and cadmium, are significant environmental contaminants frequently monitored. The Agency for Toxic Substances and Disease Registry (ATSDR) classifies these substances as high-priority pollutants due to their health risks, environmental persistence, and potential for bioaccumulation[5,6]. Given their extensive application, ensuring the reliability of urinary biomarkers is essential for accurate assessments of toxin exposure.

Twenty-four-hour urine collection is the gold standard in biomonitoring, suggesting improved reliability by minimizing the impact of diurnal fluctuations inherent to spot samples. However, it is resource-intensive, prone to contamination, and demands high participant compliance, often resulting in invalid or incomplete samples[7–9]. Spot urine sampling is more practical for clinical and epidemiological studies, but it introduces variability, as hydration affects urinary flow rates and, in turn, biomarker concentrations. Their primary dependence on urinary output and dilution (commonly referred to as diuresis) significantly undermines the reliability of uncorrected spot urinary biomarkers as indicators of exposure[3,10]. Although the need for diuresis correction in spot urine sampling is well recognized, the debate about the most accurate adjustment method persists[3,11–13].

While adjusting for urinary flow rate (UFR) is often recommended as the most accurate correction method, it is time-consuming and impractical for routine clinical and epidemiological applications[11–14]. Consequently, surrogate correctors such as creatinine (CRN), osmolality, and specific gravity are widely used, as they are thought to be similarly influenced by diuresis, like most analytes of interest[3,10,12,13]. However, recent research has revealed significant discrepancies between the hydration dependence of most analytes and standard correctors like CRN. These discrepancies can result in substantial nonlinear biases, referred to as Systemic Dilution Adjustment Errors (SDAE)[3,13]. Further challenges arise from biochemical interactions between correctors and analytes, potentially introducing biases through confounding, mediator, or collider effects[3,11,13,15,16]. The inherent limitations of dilution correction are further elaborated in *Supplementary File 1: Dilution Adjustment Errors* and, with specific reference to arsenic, in *Supplementary File 2: Arsenic Background Information*. Compensating for these nonlinear biases by a progression of previous nonlinear adjustment approaches is the primary focus of this paper[3,11,13,15,16].

Limitations due to variability in CRN excretion

Creatinine (CRN) correction is the most widely used method for dilution adjustment due to its practicality, routine measurement, and extensive supporting research[3,10,12,17,18]. In conventional CRN correction (CCRC), analyte concentrations are normalized by dividing them by CRN levels, converting the units from analyte per volume (e.g., μg/L) to analyte per creatinine weight (e.g., μg analyte/g CRN).

While CCRC helps standardize results, its accuracy is fundamentally constrained by three flawed assumptions: (1) CRN production and excretion are stable both across individuals and within the same individual over time;[3,21] (2) there are no biochemical interactions that distort the relationship between the analyte and CRN;[10,22] and (3) the urinary mass ratio of CRN to analytes remains constant across varying levels of diuresis and urine dilution[3,11]. However, all three assumptions are untenable across various populations and analytes.

Urinary creatinine (CRN) levels are primarily influenced by hydration status, alongside various interrelated factors that affect CRN production and renal excretion, as detailed in Table 1. These factors include muscle mass, dietary habits, age, sex, and kidney function. Fluctuations in CRN excretion, whether present as group-level biases (e.g., sex-based differences) or as individual variability (e.g., arising from diet, exercise, or medication), can shift corrected values upward or downward, thereby challenging CRN's role as a reliable dilution corrector[3,17,18].

While alternatives such as osmolality and its proxy, specific gravity, are less influenced by muscle mass and metabolic factors, they remain susceptible to fluctuations in glucose, protein, and electrolyte levels, particularly affecting clinical populations[34–37]. In addition, nonlinear SDAEs, like those outlined below for CRN, constrain the reliability of osmolality and specific gravity[3].

Lower urinary CRN	Higher urinary CRN
Lean muscle mass ↓, BMI ↑ Female (CRN production/kg BW: 20%↓, urine production/kg BW: 25%↑ → CRN 36%↓) Older age (constant decline of CRN after peak values in early adulthood mainly due to loss of lean muscle, malnutrition, progressive renal failure, and diuretic medication) Lack of exercise (breakdown of creatine phosphate to CRN↓) Pregnancy – 3rd term (CRN clearance↓ and CRN↓; physiological decline of CRN and rise in serum CRN until delivery date) Deficiency of methylation-associated nutrients such as protein, choline, folate, or vitamin B12 At night: CRN production & excretion rate↓ but diuresis↓↓ → CRN in morning urine↑ Diabetes mellitus and insipidus (frequently serum CRN↑) Progressive muscle dystrophies (CRN↓ is a good predictor alongside urinary creatine↑) Multifactorial influence of kidney disease: Creatine synthesis↓ due to renal cellular damage and dietary reduction of protein CRN-washout↑ due to dialysis, diuretics, or polyuria/hyperfiltration in early stages of renal insufficiency Filtration and tubular secretion of CRN↓ in advanced stages or acute renal failure	Lean muscle mass↑, BMI↓ Male Younger age (Juvenile muscle growth is accompanied by continuous increase in CRN from birth to 19–25 years of age) High fitness level, Training before analysis (especially anaerobic) Pregnancy – 2nd term (CRN clearance + 50%, CRN↑) High consumption of protein, meat, fish, or creatine, adequate supply of methylation-associated nutrients Daytime: CRN production/excretion rate↑ but diuresis↑↑ → CRN daytime urine↓ Medication (statins, some antibiotics and antacids) Muscle trauma, rhabdomyolysis, cramps

Table 1. Physiological, pathological, and epidemiological influences on CRN. Urinary dilution is the most significant determinant of CRN. The other listed factors modify its impact[7,8,12,19–33].

More advanced covariate adjustment models have been developed in recent years to account for confounding variables affecting CRN and analytes[10,12,38,39]. However, their widespread adoption in routine biomonitoring has been limited by logistical challenges and the complexity of their application. Additionally, multi-linear covariate models can introduce collinearity issues, particularly when covariates are highly intercorrelated, as is often the case with factors like creatinine, muscle mass, BMI, age, and sex[38]. Moreover, nonlinear behavior in these covariates can further undermine the robustness of the model[40].

Distortion due to systemic dilution adjustment errors (SDAEs)

Beyond the imprecision caused by uncontrolled variation in creatinine (CRN) excretion, conventional CRN correction (CCRC) is further limited by nonlinear biases[3,11,13,41,42]. Figure 1 illustrates the differing responses of CRN and other analytes to hydration changes, leading to systemic dilution adjustment errors (SDAE) that vary by analyte. A primary source of this discrepancy lies in the unique excretion behavior of CRN, which is predominantly glomerular filtered with minimal tubular reabsorption[43–45]. In contrast, many analytes deviate significantly from CRN in renal excretion, as their tubular net secretion rates fluctuate more with urinary flow rate than those for CRN[46,47]. This discrepancy introduces a nonlinear dynamic, causing analyte levels to be overestimated at higher and underestimated at lower dilutions[3,11,13,41,42]. The biochemical complexities underlying these dynamics are further elaborated in *Supplementary File 1* and, specifically for arsenic, in Section 5 of *Supplementary File 2*.

Given these challenges, the World Health Organization (WHO) recommends restricting acceptable dilution ranges to mitigate the variability introduced by extreme dilution levels[48]. According to WHO guidelines, "it is unlikely that any correction will give accurate results when urine is either very dilute (relative density < 1.010 or CRN < 0.3 g/L [< 2.65 mmol/L]) or very concentrated (relative density > 1.030 or CRN > 3.0 g/L [> 26.5 mmol/L])."[48]. Notably these restrictions apply not only to CRN but also to specific gravity (relative density).

While limiting dilution ranges minimizes errors from extreme samples, this approach excludes a substantial portion of the dataset. For instance, in our iodine dataset (*Excel File* 1, Sheets 16 and 17), applying the CRN restriction of 0.3–3 g/L rejects 19.9% of samples, including 21.4% from females and 10.9% from males. Sample rejections disproportionately affect individuals with higher urine dilution levels, such as women, seniors, and children, who naturally exhibit lower CRN production and excretion. Excessive fluid intake and intravenous infusions, which result in overly diluted samples, are also common reasons for sample rejection[24].

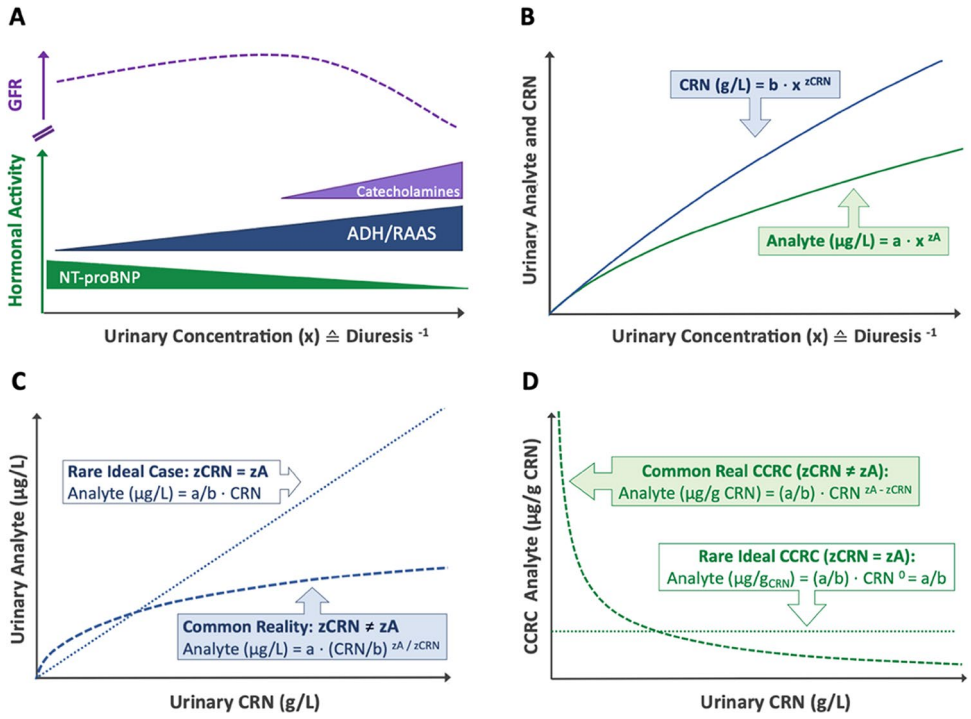


Fig. 1. Origin of the systemic dilution adjustment error (SDAE) illustrated by the differential dependence of urinary levels of CRN and analyte on diuresis in a fictitious subject under assuming constant CRN production and analyte exposure. (**A**) Relationship between urine concentration, glomerular filtration rate (GFR), and significant fluid-regulating hormone activities. (**B**) Dependence of CRN and urinary analyte concentrations on diuresis as represented by uncorrected and conventionally CRN corrected (CCRC) form. (**C**) Ideal and natural relations between uncorrected analyte (µg/L) and CRN. (**D**) Ideal vs. actual relations between CCRC analyte (µg/g CRN) and CRN. Abbreviations: RAAS = renin-angiotensin-aldosterone system, ADH = antidiuretic hormone/vasopressin, BNP = brain natriuretic peptide.

These limiting practices strain resources, diminish statistical power, and risk introducing selection bias by ultimately underrepresenting key populations.

The distortions caused by unaccounted nonlinear SDAEs extend beyond CRN adjustments to other methods, such as osmolality and specific gravity[39]. Even advanced multiple linear regression models, which similarly restrict dilution ranges, fail to capture the actual nonlinear dynamics of analyte excretion[39,49]. Excluding extreme samples may reduce nonlinear effects but not eliminate persistent inaccuracies within accepted ranges, ultimately skewing outcomes in clinical, epidemiological, and forensic research. Moreover, the lack of consistent physiological or statistical justifications for restricting dilution ranges has led to variability across laboratory standards[50]. To address these challenges, advanced nonlinear adjustments—such as power-functional, log-linear, and polynomial methods—are increasingly being explored as robust alternatives to conventional approaches[3,11,13,41,42].

Advancements in urinary biomarker adjustment through nonlinear models

Advancements in urinary biomarker adjustment have focused on refining correction methods by normalizing to fixed values, such as 1 ml/min UFR or 1 g/L CRN[3,11]. These approaches have shown promise in reducing the residual dependency of results on urinary dilution and enhancing correlations with external exposure variables[3,11,13,15,16,41,42].

In a recent study, Middleton et al. (2019) evaluated a power-functional adjustment method for urinary arsenic levels, comparing it to conventional CRN correction (CCRC), osmolality, and specific gravity[3]. The method normalized analyte levels to 1 g/L CRN using the formula:

$$\text{Analyte Normalized to 1 g/L CRN} = \text{Analyte Uncorrected}/\text{CRN}^{0.8} \tag{1}$$

This adjustment reduced residual dependencies of conventionally corrected (CCRC) arsenic results on diuresis and enhanced correlations with arsenic levels in drinking water. The approach also proved effective for osmolality and specific gravity, which were similarly normalized in a nonlinear manner. Systematic testing of various exponents **b** identified **b** = 0.8 as optimal (see Eq. 1 and Fig. 1)[16]. Despite a limited sample size of 202 spot urine samples, which restricted the ability to assess confounding factors such as sex, age, or muscle mass, the study demonstrated significant improvements. Notably, their power-functional approach mitigated even some of the existing limitations of CRN compared to osmolality and specific gravity in traditional adjustments by reducing dilution bias[3]. These findings highlight the potential of nonlinear adjustments to advance urinary biomarker research and provide more reliable results in clinical and epidemiological applications.

Introducing variable power functional CRN correction (V-PFCRC)

Failing to account for the differential renal handling of correctors (CRN) and analytes during diuresis variability, traditional and even advanced nonlinear or covariate models often remain prone to nonlinear biases and overcorrection, as illustrated in Figure 1. These limitations are further compounded by exposure-dependent fluctuations in analyte excretion (Figs. 1, 3 and 4; Supplementary Figs. 1 and 2)[3,13,16,25,39]. To address these challenges, the Variable Power-Functional CRN Correction (V-PFCRC) method was developed to dynamically estimate the expected analyte value at a CRN concentration of 1 g/L for any analyte-CRN combination.

Unlike traditional methods relying on fixed exponents (e.g. S-PFCRC) or linear assumptions, the V-PFCRC dynamically adjusts the exponent in the correction formula based on exposure level. This adaptive approach enables more precise and nuanced correction of nonlinear systemic dilution adjustment errors (SDAEs). The method integrates two empirically derived and physiologically based mathematical principles through a dual power-functional/log-linear regression process, facilitating the conversion of raw analyte percentiles into normalized values at a CRN concentration of 1 g/L.

The V-PFCRC compensates for exposure-dependent variations more effectively than fixed-exponent power-functional models or traditional methods by varying the correction exponent. Its efficacy has been rigorously validated across multiple datasets, demonstrating significant reductions in SDAEs and improved accuracy in normalized analyte values[51]. Validation studies included arsenic, four other metals, and iodine as a representative non-metal, highlighting the method's broad applicability. The development, mathematical principles, and formula derivations of V-PFCRC are detailed in *Supplementary File 3: V-PFCRC Formula Determination*.

Considerations in marker selection for initial development

For the initial development of the Variable Power-Functional CRN Correction (V-PFCRC) method, total urinary arsenic (TWuAs) was chosen as the primary marker due to its relevance as an environmental toxin primarily excreted through the kidneys[13,35]. Arsenic can be reliably quantified in urine samples above 1 µg/L using Inductively Coupled Plasma Mass Spectrometry (ICP-MS), a widely accepted standard for arsenic analysis[52]. While TWuAs lacks specificity in distinguishing between arsenic species, it remains the standard biomarker for initial clinical assessments, leveraging a robust dataset of 5,553 spot urine samples from IMD Berlin, which formed the foundation for the first numeric evaluation.

Urinary biomarkers such as arsenic, iodine, and zinc often encompass diverse chemical forms derived from multiple sources, providing a broad exposure overview. However, this heterogeneity complicates the attribution of specific sources and associated toxicological risks. Additionally, the renal excretion profile of a biomarker may vary with its concentration, as exposure levels influence chemical composition and excretion dynamics. For arsenic, this variability is particularly evident in regions with low baseline levels of inorganic arsenic in drinking water, such as Central Europe, where compliance with the WHO threshold of 10 µg/L is generally high[53–55].

In these contexts, elevated TWuAs levels are often associated with non-toxic organic arsenic species, such as arsenobetaine, from seafood consumption, which can raise urinary arsenic concentrations to over 800 µg/L. In

contrast, individuals without such exposure typically exhibit urinary arsenic levels between 5 and 50 µg/L[56,57]. Table 2 overviews the primary arsenic species in human urine and blood, their sources, and toxicity profiles. Detailed arsenic speciation, such as inorganic arsenic, monomethylarsonic acid (MMA), and dimethyl-arsinic acid (DMA), is usually reserved for significantly elevated urinary arsenic samples.

Initially developed using Arsenic, the V-PFCRC method could be extended to urinary iodine and other metals, including cesium, molybdenum, strontium, and zinc, demonstrating its versatility across a broader range of biomarkers. By overcoming the limitations of conventional CRN correction, the V-PFCRC method offers a reliable and adaptable tool for improving the accuracy of urinary biomarker analyses. The methodology behind its development and application is presented in the following section.

Study objectives

1. **Develop a Mathematical Standardization Method**.
2. **Create a refined, user-friendly post-analytical method for standardizing analytes to urinary CRN at 1 g/L**, addressing nonlinear biases across varying exposure levels.
3. **Define Valid Analyte and CRN Ranges**.
4. **Investigate computational behavior in extreme ranges** to establish valid concentration thresholds.
5. **Validate Standardization Function**.
6. **Assess residual dependencies in aggregate and sex-disaggregated data and correlate total urinary and blood arsenic levels in parallel measurements.**
7. **Evaluate Applicability**.
8. **Test the generalizability of the V-PFCRC method across additional elements**.
9. **Assess Confounder Impact**.
10. **Analyze V-PFCRC validity in hypothetical populations with varied CRN excretion rates**, focusing on the influence of CRN-related confounders.

Methodology

Study population

An anonymized dataset of 5,752 spot urine samples (Set 1), collected between January 2014 and August 2022 by the Institute for Medical Diagnostics Berlin-Potsdam (IMD), served as the basis for developing the V-PFCRC method. This dataset included age, sex, creatinine (CRN) levels, and total weight urinary arsenic (TWuAs). Samples were obtained from routine multi-element screenings across Germany, Switzerland, and Austria as part of preventive diagnostics, resulting in a demographically diverse dataset. Given the generally low background exposure to inorganic arsenic in these regions, it was assumed that there was no specific clinical indication or suspicion of arsenic-related toxicity, guiding the multielement spot urinary analysis[53,54,56,57].

A secondary dataset of 1,175 samples (Set 2), collected between April 2014 and March 2024, included parallel measurements of TWuAs and EDTA blood arsenic.

Restricting the age range to 14–82 years resulted in 1,145 remaining samples, 812 of which overlapped with Set 1 and 333 of which were newly collected.

To assess the generalizability of V-PFCRC beyond arsenic, analyses were extended to cesium ($n = 3{,}500$), strontium ($n = 3{,}560$), molybdenum ($n = 4{,}044$), zinc ($n = 4{,}025$), and a large dataset of 58,439 iodine samples (females: 49,745; males: 8,694). IMD also provided parallel blood measurements for 1,823 iodine samples. Raw data and basic statistics for all six elements analyzed are available in *Excel File* 1: *Data Sets*.

This retrospective study did not include speciation data or information on diagnoses, indications, medications, or specific exposure sources (e.g., diet or water), as these details were not essential to its primary objective. The study aimed to develop a generalized corrective mathematical framework based on six urinary biomarkers, including arsenic. Within this context, incorporating arsenic speciation was deemed unlikely to add substantial value to the model, as its contribution to the overall framework was expected to be minimal. The large, diverse sample set helps average out individual variability in CRN production, arsenic methylation, and urinary species composition, enabling a more accurate assessment of the physio-mathematical underpinnings of systemic dilution adjustment errors (SDAE) and their correction by V-PFCRC.

Trimming of data – exclusions based on CRN, age, and arsenic

Data and key statistical details for all trimmed sample sets are summarized in *Excel File* 1.

Exclusions based on CRN levels

To reduce variability, six creatinine (CRN) levels exceeding 4.5 g/L were excluded from the initial 5,752 arsenic samples.

Age-based exclusions

Children under 14 years were excluded due to their distinct toxicokinetics of arsenic methylation and excretion and rapid changes in CRN metabolism associated with muscle growth. This led to the exclusion of 151 samples (42 females, 109 males). Additionally, 42 subjects above 82 years (19 females, 23 males) were excluded due to potential confounding factors such as comorbidities, chronic kidney disease, and complex medication regimens, which could influence urinary arsenic and CRN levels.

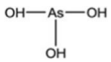
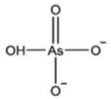
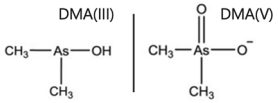
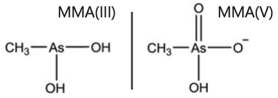
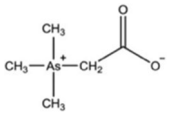
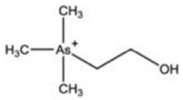
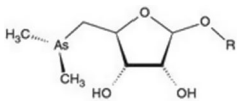
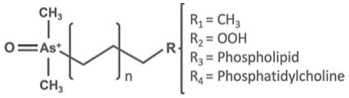
Name	Abbr.	Chemical Structure	Occurrence/Origin[56,68,69]	Toxicity[70,71]	% of TWuAs[49,72,73]	% in blood As[49]	pKa[74]
Arsenite (inorganic)	As(III) iAs(III)		Drinking water, Rice, Flour, various foods, Pesticides	+++	1–3%	20.2%	$pKa_1 = 9.23$ $pKa_2 = 12.13$ $pKa_3 = 13.4$
Arsenate (inorganic)	As(V) iAs(V)		Drinking water, Rice, Flour, various foods, Pesticides	++	0.1–0.3%		$pKa_1 = 2.19$ $pKa_2 = 6.98$ $pKa_3 = 11.53$
Dimethylarsinic Acid	DMA		Seafood (small amounts), Primary metabolite of iAs, Arsenosugars, and -lipids	DMA(III)++++ DMA (V)+	30–41%	11.7%	pKa = 6.2
Monomethyl-arsonic Acid	MMA		Seafood (small amounts) Metabolite of iAs, Arsenosugars, and -lipids	MMA(III)+++++ MMA(V)+	2–5%	2.4%	$pKa_1 = 4.1$ $pKa_2 = 8.7$
Arsenobetaine (75–100% of 'Fish-arsenic')	AsB		Mollusks, Crustaceans, Cephalopods, Finfish, Shellfish, Demersal Fish	(+)?	55–65%	51.1%	pKa = 2.18
Arsenocholine	AsC		Seafood, Fish, Fish oils	(+)?	<1%	4.5%	
Arsenosugars	AsS		Major As compound in seaweed (brown kelp). Present in mussels, oysters, clamps	(+)?	Metabolized by humans mainly into DMA[69]		
Arsenolipids 1. As-containing Hydrocarbons 2. As-containing Fatty Acids 3. As-containing Phospholipids 4. As-containing Phosphatidylcholines	1. AsHCs 2. AsFAs 3. AsPLs 4. AsPCs		1. Fish oils 2. Fish oils 3. Seaweed 4. Roe	(+++)? (+)? (+)? (+)?	Metabolized by humans mainly into DMA[70]		

Table 2. Compilation of the most abundant as species found in humans.

Outliers in arsenic values
After trimming for CRN and age, 5,553 samples (3,024 females and 2,529 males) were left for V-PFCRC analysis. To minimize the impact of arsenic outliers, the upper 1% of arsenic values within each CRN band were excluded when generating corrective formulas, as detailed in *Supplementary File 3*.

Treatment of arsenic values below the detection limit

The study included 116 female and 32 male urine samples with urinary arsenic levels below the 1 µg/L detection limit of the ICP-MS method. A distribution between 0.5 and 0.99 µg/L was assumed for these samples, as lower values seemed unlikely due to the prevalent arsenic background exposure. To approximate the values below the detection limit, the dependence of uAs_{UC} on urinary concentration was considered using the formulas:

$$\text{Females estimated } \mathrm{uAs_{UC}} = 0.92 + 0.17 \times \ln(\text{CRN g/L})$$

$$\text{Males estimated } \mathrm{uAs_{UC}} = 0.99 + 0.20 \times \ln(\text{CRN g/L})$$

This way, the samples below the detection limit were distributed in the 0.5–0.99 µg/L range, depending on CRN, with average estimated arsenic values of 0.62 µg/L for females and 0.72 µg/L for males. This estimation approach produced significantly more plausible value distributions compared to omitting values or using common fixed substitutions, such as replacing values below the detection limit with the detection limit itself (1 µg/L), half of it, or divided by the square root of 2 (0.71 µg/L). It should be noted that the different handling of values below the detection limit does not cause any essential variability in the correction formulas and their efficiency. Similar calculations were performed for the 69 blood samples below the detection limit of 0.2 µg/L accounting for the correlation between arsenic in blood and urine using the unisex formula:

$$\text{Estimated blood As} = 0.082 + 0.028 \times \ln \ (= \mathrm{uAs}_N \text{in } \mu\text{g/1g CRN})$$

This estimate distributed the blood values below the detection limit between 0.08 and 0.19 µg/L with an average value of 0.12 µg/L corresponding to 60% of the detection limit.

Samples collection, preservation, transportation, and storage procedures

Urine: Urine samples were collected in neutral monovettes (Sarstedt) and shipped to the laboratory at room temperature within 24 h for immediate CRN measurement. They were then stored at 4 °C for 1–4 days for subsequent ICP-MS measurement.

Blood: Venous whole blood was collected in sampling tubes with and without EDTA or heparin for anticoagulation (Sarstedt) and sent to the laboratory by overnight transport or surface mail. The serum was separated from whole blood samples without EDTA/heparin anticoagulation by centrifugation at 2000 x g for 10 min. The stability of elements in human specimens was validated over 14 days at room temperature.

Sample preparation and analysis procedures

Inductively coupled plasma-mass spectrometry (ICP-MS) measurements of metals
Metal ion concentrations were determined by inductively coupled plasma-mass spectrometry (ICP-MS) in serum (iodine), anticoagulated whole blood (total weight arsenic), and urine (total weight arsenic, cesium, iodine, molybdenum, strontium, zinc). Arsenic species were not analyzed. In preparation for ICP-MS analysis, Urine was diluted 1:10 in 1% HNO_3 (Suprapur, Supelco), and serum and anticoagulated whole blood samples were diluted 1:20 in high purity 0,1% NH_3 (Suprapur, Supelco) / 0,02% Lutrol F88 (AppliChem). Metals were analyzed in collision/reaction cell mode by ICP-MS (ICapQ, Thermo Fisher), using external and internal standard calibration (Elemental Scientific). Results represent means of triple detections.

Creatinine measurement
Creatinine analysis was performed enzymatically using the Alinity assay (Abbot Laboratories).

Derivation of V-PFCRC formulas

The Variable Power-Functional CRN Correction (V-PFCRC) formulas were developed to address exposure and dilution effects. Key steps involved dividing creatinine (CRN) values into intervals, ranking uncorrected analyte values by percentiles within each CRN band, and applying regression analysis to identify a power-functional relationship (Fig. 2; Supplementary Fig. 3). A second regression established a log-linear link between the exponent **b** and the normalized analyte level, reflecting exposure. Combining these two steps produced sex- and analyte-specific coefficients, **c** and **d**, which define the corrective V-PFCRC formula:

$$A_{\mathrm{N}} = A_{\mathrm{UC}}/\mathrm{CRN}^{[(c \cdot \ln A_{\mathrm{UC}} + d)/(c \cdot \ln \mathrm{CRN} + 1)]}$$

$$A_N : \text{Analyte normalized to } \mu\text{g/1 g CRN}$$

$$\mathrm{A_{UC}} : \text{ Uncorrected analyte concentration}$$

For more comprehensive details on formula derivations, refer to *Supplementary File 3*.

Accuracy of the V-PFCRC correction across various CRN cutoffs
The correlation strength in both regression steps, as indicated by R^2 values, reflects the effectiveness of the V-PFCRC correction and varies depending on the element and CRN cutoffs used. A systematic analysis of the impact of CRN cutoffs from 2.4 to 4.4 g/L on curve fitting and the accuracy of V-PFCRC is detailed for all elements and both sexes of *Excel File 2*: V-PFCRC Formula Validation, Sheets 3–5.

Compensating asymmetrical CRN distribution of datasets
CRN values exhibit an asymmetric distribution (*Excel File 1*, Sheet 4, B), potentially impacting generalizability and increasing the risk of overfitting. To address this, uncorrected arsenic and CRN values were standardized by

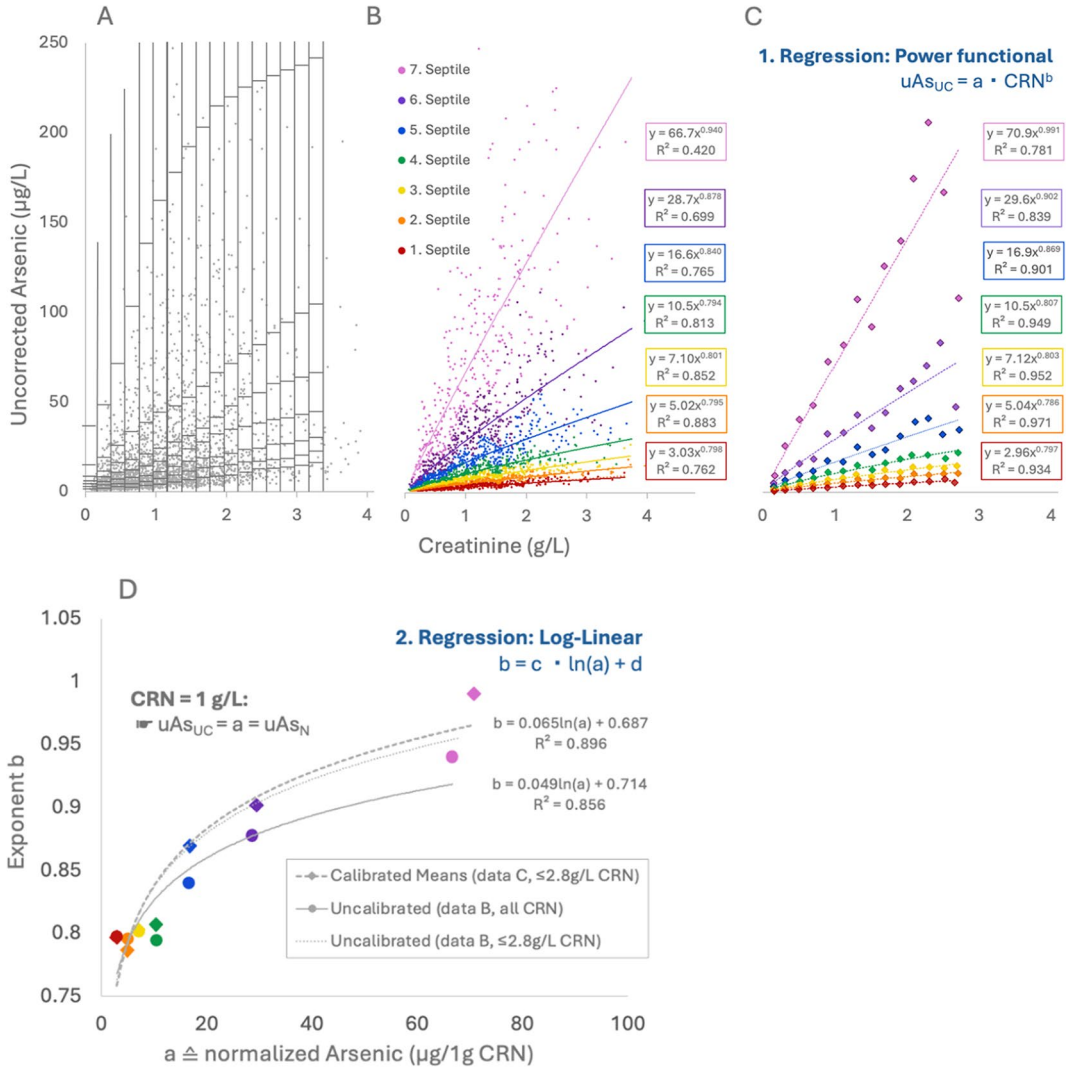


Fig. 2. The Four Steps of Numerical V-PFCRC Formulas Determination. (**A**) Percentile ranking of uAsUC concentrations across 17 CRN bands (width: 0.2 g/L CRN). (**B**) Filtering of uAsUC septiles, excluding the top 1% within each band, and power-functional regression for each septile. (**C**) Calibration of asymmetric CRN distribution by averaging septiles within each band, enabling symmetrical mapping of the diuresis range (≤ 2.4 ≤ 2.4 g/L for females, ≤ 2.8 ≤ 2.8 g/L for males). (**D**) Log-linear regression of arsenic normalized to 1 g/L CRN (uAsN) and the corresponding exponent **b**, yielding analyte-specific coefficients **c** and **d**.

calculating the septile mean for each CRN band (Fig. 2c) and performing regression on these means rather than all individual data points.

This approach improved coverage of previously underrepresented CRN regions. The effects of CRN-asymmetry compensation on coefficients, dilution bias reduction, and blood-urine correlation improvement were assessed (Fig. 2d; Supplementary Fig. 4).

Definition of 14 result modes

Coefficients **c** and **d** were derived under varying conditions, including sex-differentiated vs. aggregated data, CRN asymmetry-compensated vs. uncompensated models, and two datasets (Set 1 and Set 2). This process resulted in six V-PFCRC modes, each with subtle differences in coefficients **c** and **d**, to be assessed for their effectiveness in reducing dilution bias and improving blood-urine correlations. An additional six analogous S-PFCRC modes were included, where the fixed exponent **b** was calculated as the average of the **b** values from power-functional curves generated during the first regression step. Combined with uncorrected and CCRC modes, this yielded 14 distinct result modes for comparison, as defined in Supplementary Table 1.

Statistical analysis

Software

Data were analyzed using Microsoft Excel, Datatab.net, and Python scripts executed in Jupyter Notebooks for formula generation and CRN cutoff validation. A summary of the Python/Jupyter results, including the scripts

used, is provided in *Excel File* 2. These scripts accurately reproduce the analysis of arsenic Set 1 and can be adapted for V-PFCRC analysis of other datasets and biomarkers.

Data normalization and statistical testing
Log transformation was applied to approximate normality. Data distribution was assessed using Q-Q plots (*Excel File* 1, Sheet 4) and Kolmogorov–Smirnov tests. Results are expressed as mean ± standard error, medians, and quartiles (Q1, Q3). For normal or log-normal data, Student's t-tests and Pearson correlations were used; for non-normal data, Spearman correlations and Mann–Whitney U-tests, with $p < 0.05$ considered significant.

Intercompare arsenic results modes in complete data sets
Key statistics (means, SD, quartiles, IQR, skewness, and kurtosis) were calculated for females and males and combined data across 14 modes for the entire dataset and a subset with CRN ≤ 2.4 g/L. Spearman correlation was used to assess the relationships (r, p) between arsenic levels, age, and CRN for both sexes and combined data. Sex-differentiated V-PFCRC values (Mode 5) were compared to uncorrected (Mode 1), CCRC (Mode 2), sex-aggregated V-PFCRC (Mode 6), and sex-differentiated S-PFCRC (Mode 9) using Bland-Altman plots and scatter diagrams (Supplementary Fig. 5) to identify differences and potential outliers.

Intercompare different arsenic results modes in exposure septile stratified data
To evaluate the impact of arsenic exposure levels on correction efficiency, the residual dependence of log-transformed V-PFCRC values (mode 5) on CRN was analyzed through regression for each arsenic septile and compared to uncorrected, CCRC and sex-differentiated S-PFCRC. Residual CRN bias was assessed across female, male, and combined datasets using arsenic slopes and R^2 values. Python codes for the analysis are included in *Excel File* 2, Sheet 8.

Correlation of arsenic levels in urine and blood
The interpretation of these correlations is complicated due to blood arsenic levels often being near detection limits, reducing resolution. Urinary medians were 13.5 times higher than those in blood[55,56,58–61]. This discrepancy can lead to significant measurement errors at low blood concentrations. Therefore, additional Spearman correlations were performed on subsets excluding samples below specific thresholds (quartiles 1, 2, 3, or the 90th percentile).

V-PFCRC of other elements
Analogous assessments were conducted for iodine as a non-metal and four additional metals to evaluate the generalizability of V-PFCRC principles beyond arsenic. This included extensive datasets for urinary iodine (n = 58,439; F = 49,745; M = 8,694) and substantial datasets for cesium (n = 3,411; F = 1,850; M = 1,561), molybdenum (n = 3,913; F = 2,138; M = 1,775), strontium (n = 3,380; F = 1,813; M = 1,567), and zinc (n = 3,917; F = 2,178; M = 1,739). Age was restricted to 14–82 years for all elements. The same Python scripts used for arsenic were applied, with results detailed in *Excel File* 2, Sheets 3–5. For a subset of 1,823 samples (F = 1,540; M = 283), parallel EDTA blood measurements were included to examine correlations between urine correction modes and blood levels.

Exploratory analysis of V-PFCRC formula stability

Stability analysis of V-PFCRC adjustments across parameter variations
The stability of the V-PFCRC formula for arsenic (mode 5) was evaluated across a wide range of hypothetical parameter variations to assess its robustness. Control calculations were conducted using Python/Jupyter and visualized across 14 hypothetical creatinine (CRN) concentration curves (0.001–7.5 g/L) and urinary arsenic uncorrected concentrations (uAsUC) ranging from 0 to 10,000 µg/L. These ranges were selected to simulate extreme dilution and exposure spectra, which are uncommon in real-world scenarios. Two empirically derived V-PFCRC formulas, differing in the sign of coefficient **c**, were evaluated. The first formula, with a positive **c**, was from male arsenic data with a CRN cutoff of ≤ 2.4 g/L, and the second formula, with a negative **c**, restricted data to ≤ 3.8 g/L. The derivation details for both formulas are provided in *Excel File* 2, Sheet 3. Control calculations are visualized in Fig. 7.

Assessing susceptibility of V-PFCRC and CCRC to random CRN excretion confounders
V-PFCRC and CCRC formulas were evaluated for robustness against intra- and inter-individual variations in creatinine (CRN) excretion using Python/Jupyter. The impact of potential variations in CRN excretion due to confounding factors (Table 1) on result stability was analyzed across the CRN range for both sexes at fixed urinary arsenic concentrations of 3, 10, or 40 µg/L. Details of the analysis are provided in Excel File 2, Sheet 6.

V-PFCRC compensation for systemic differences in CRN-levels
A control analysis evaluated the ability of V-PFCRC formulas to account for systemic differences in CRN production and excretion across populations (e.g., men vs. women, vegetarians vs. carnivores). CRN values in the original arsenic dataset (Set 1) were adjusted by ± 25%, and V-PFCRC formulas were recalculated for the original, + 25%, and − 25% CRN levels across all seven exposure septiles. Results at 1 g/L CRN (coefficient a for V-PFCRC) and similar analyses for uncorrected and CCRC arsenic values were compared in *Excel File* 2, Sheet 7.

Effects of age-normalized CRN adjustments on V-PFCRC formulas
Gender-specific polynomial formulas were developed using CRN data from the extensive iodine dataset to evaluate the impact of age normalization on the V-PFCRC method (*Excel File* 1, Sheet 15). A gender-specific

age correction factor F_{AGE} was calculated by dividing the expected mean CRN value for a 50-year-old by the expected mean CRN for the individual's actual age:

$$F_{AGE} = \text{Mean CRN Age 50 (years)}/\text{Mean CRN Patient Age (years)}$$

This factor was then applied to adjust CRN levels:

$$\text{CRN}_{\text{AGE-adjusted}} = \text{CRN} \times F_{AGE}$$

V-PFCRC formula calculations were then performed using $\text{CRN}_{\text{AGE-adjusted}}$ instead of the original CRN. The validity of the V-PFCRC principle for these data sets was assessed based on curve fits across a wide range of CRN cutoffs as plotted in *Excel File* 2, Sheets 3 and 5.

Results

V-PFCRC formulas for arsenic and other elements

The power functional curves for each septile in the initial regression step and the log-linear relationship between exponent **b** and arsenic normalized to 1 g/L CRN are shown in Fig. 3 for women and Supplementary Fig. 1 for men. Using CRN cutoffs of ≤ 3.0 g/L for women (99.1% of samples) and ≤ 3.8 g/L for men (99.6% of samples), both sexes demonstrated strong curve fits in both regression steps. Similar patterns were observed for other analyzed elements (*Excel File* 2, Sheets 3–5). Supplementary Fig. 6 demonstrates the robust log-linear relationship between **a** and **b** for the extensive iodine dataset, highlighting a fundamental principle underlying the V-PFCRC formulas.

Validation of V-PFCRC dilution adjustment efficiency for arsenic

To validate the efficacy of V-PFCRC dilution adjustment across datasets covering all exposure levels, basic statistics of the ten arsenic result modes for Set 1 (Supplementary Table 1, modes 1–10) were intercompared (*Excel File* 1, Sheet 5, A–D) and residual correlations with CRN were evaluated (Table 3; *Excel File* 1, Sheet 5, E). Correlations between urine and blood in Set 2 were analyzed across all 14 arsenic result modes (Table 4; *Excel File* 1, Sheet 6). Residual biases by exposure level are shown in Figs. 3 and 4, and Supplementary Fig. 1.

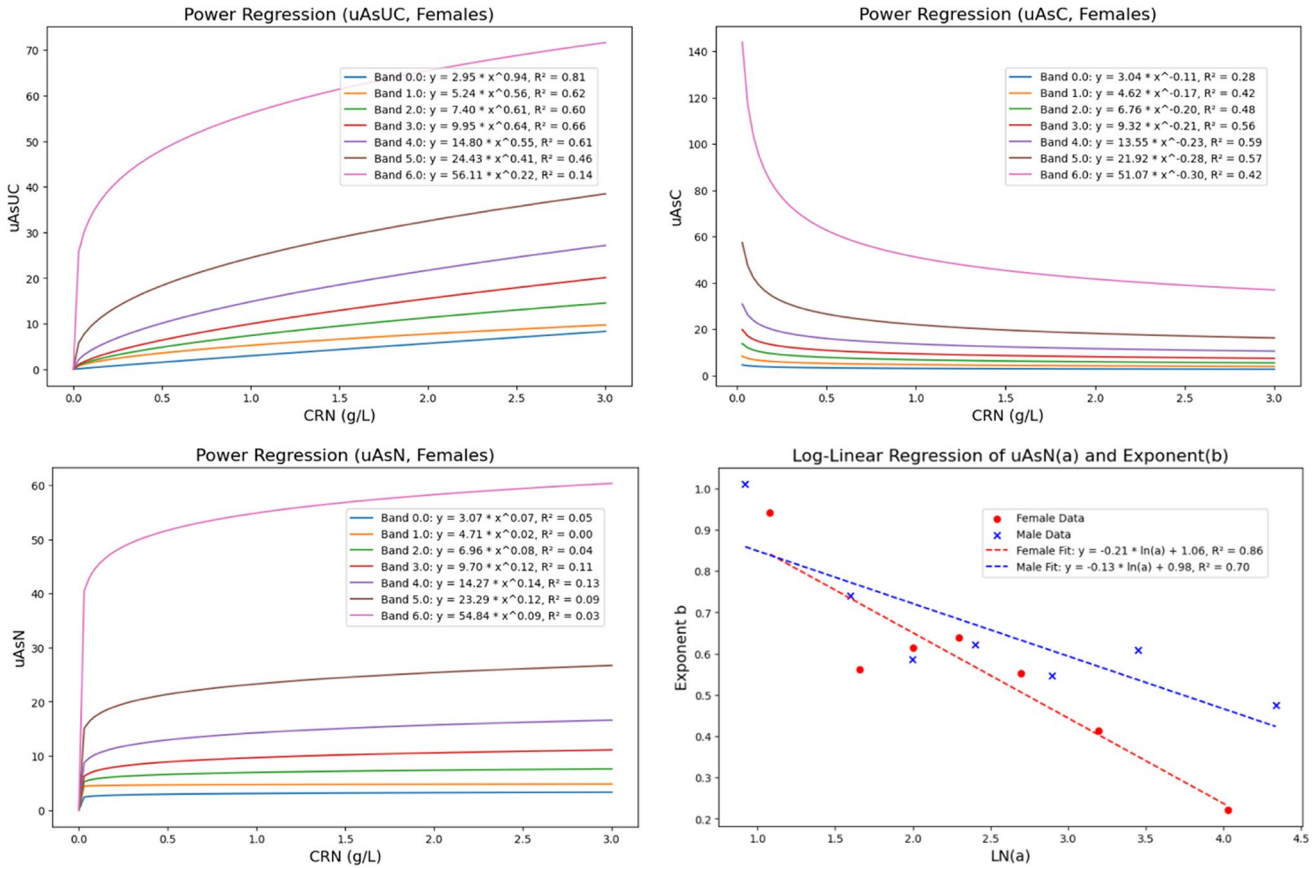


Fig. 3. Residual dependency on CRN for females using V-PFCRC, uncorrected, and CCRC arsenic levels, stratified by exposure categories. The first power-functional regression step is plotted for all seven exposure levels, with curve fits and corresponding coefficients (**a**) and exponents (**b**) for uncorrected, CCRC, and V-PFCRC (mode 5) arsenic. The bottom right graph illustrates the robust log-linear relationships between a and b for both sexes. Analysis was conducted using Python/Jupyter with a CRN cutoff of ≤ 3.4 g/L, including the 0–99th percentiles for each CRN band. Corresponding male data are shown in Supplementary Fig. 1.

All CRN		BASIC Parameters				V-PFCRC				S-PFCRC			
Result Mode Nummer				**1**	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**	**10**
Spearman CorrelationFemale, *n* = 3024		**Age**	**CRN (g/L)**	**uAs_{UC} (µg/L)**	**uAs_C (µg/g CRN)**	**S-DIF *N*-CAA All CRN**	**S-AGG *N*-CAA All CRN**	**S-DIF CAA CRN ≤ 2.4**	**S-AGG CAA CRN ≤ 2.4**	**S-DIF *N*-CAA All CRN**	**S-AGG *N*-CAA All CRN**	**S-DIF CAA CRN ≤ 2.4**	**S-AGG CAA CRN ≤ 2.4**
Age	**r**	1	-0.11	0.01	0.1	0.09	0.08	0.08	0.09	0.09	0.09	0.09	0.09
	p	-	< 0.001	0.455	< 0.001	< 0.001	< 0.001	< 0.001	< 0.001	< 0.001	< 0.001	< 0.001	< 0.001
CRN	**r**	**-0.11**	**1**	**0.56**	**-0.12**	**0**	**0.06**	**0.03**	**0**	**0.01**	**0.01**	**0.03**	**0**
	p	< 0.001	-	< 0.001	< 0.001	0.89	0.001	0.125	0.933	0.671	0.776	0.072	0.804
Male, *n* = 2529		**Age**	**CRN**	**1**	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**	**10**
Age	**r**	1	-0.14	0.08	0.15	0.14	0.13	0.14	0.14	0.14	0.14	0.14	0.14
	p	-	< 0.001	< 0.001	< 0.001	< 0.001	< 0.001	< 0.001	< 0.001	< 0.001	< 0.001	< 0.001	< 0.001
CRN	**r**	**-0.14**	**1**	**0.47**	**-0.11**	**0**	**0.03**	**-0.03**	**-0.02**	**-0.01**	**-0.01**	**-0.03**	**-0.02**
	p	< 0.001	-	< 0.001	< 0.001	0.806	0.09	0.203	0.404	0.636	0.626	0.092	0.313
All, *n* = 5553		**Age**	CRN	**1**	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**	**10**
Age	**r**	-	-0.11	0.04	0.12	0.11	0.11	0.11	0.11	0.11	0.11	0.11	0.11
	p		< 0.001	0.002	< 0.001	< 0.001	< 0.001	< 0.001	< 0.001	< 0.001	< 0.001	< 0.001	< 0.001
CRN	**r**	**-0.11**	**-**	**0.53**	**-0.12**	**0**	**0.05**	**0.01**	**-0.01**	**0**	**0**	**0.01**	**-0.01**
	p	< 0.001		< 0.001	< 0.001	0.953	< 0.001	0.578	0.586	0.915	0.986	0.647	0.421

Table 3. Spearman correlation coefficients and significances (p) between age, CRN, and urinary arsenic in complete data of the ten different result modes of sample set 1. Analog results for the CRN-restricted data to ≤ 2.4 g/L are given in *Excel file* 1, sheet 5, F. CAA CRN-asymmetry adjusted, N-CAA Non-CRN-asymmetry adjusted, S-DIF Sex-differentiated, S-AGG Sex-aggregated.

	Dilution adjustment	Result mode	Complete Set (≥ 0.08 µg/L)	≥ 1.Quartile (≤ 0.5 µg/L)	≥ 2. Quartile (≤ 0.9 µg/L)	≥ 3. Quartile (≤ 1.5 µg/L)	≥ 90th %til (≤ 3.9 µg/L)
Set	**Sample Size (*n*)**		**1145**	**800**	**551**	**280**	**114**
	None	1	0.67	0.63	0.57	0.4	0.3
	CCRC	2	0.81	0.75	0.67	0.53	0.42
Set 1	V-PFCRC	3	0.81	0.75	**0.68**	**0.54**	0.42
		4	0.8	0.75	0.67	0.53	0.41
		5	0.81	0.75	0.67	**0.55**	**0.43**
		6	0.81	0.75	**0.68**	**0.54**	0.42
	S-PFCRC	7	0.81	0.75	**0.68**	**0.54**	0.42
		8	0.81	0.75	**0.68**	**0.54**	0.42
		9	0.81	0.75	**0.68**	**0.54**	0.42
		10	0.81	0.75	**0.68**	**0.54**	0.42
Set 2	V-PFCRC	11	0.8	0.75	**0.68**	**0.54**	**0.43**
		12	0.81	0.75	**0.68**	0.53	0.42
	S-PFCRC	13	0.8	0.75	**0.68**	**0.54**	0.41
		14	0.81	0.75	**0.68**	**0.54**	0.42

Table 4. Spearman correlation r between EDTA blood arsenic and urinary arsenic of 14 result modes of table 3 in complete sex-aggregate data and after preclusion of blood arsenic values arsenic below quartile limits 1–3 or the 90th percentile. In all analyses, the correlation of urine arsenic of the 14 result modes with blood arsenic proved significant ($p > 0.001$). Bold numbers indicate improved correlations of power functional modes compared to CCRC.

Basic statistics S-PFCRC and V-PFCRC adjusted data sets

The sex-differentiated statistical parameters for uncorrected, conventionally, and power-functionally adjusted data (*Excel File* 1, Sheet 5, A, B) revealed gender-similar levels of arsenic in quartiles 1–3 across different adjustment modes. While CCRC was consistently ranked highest and uncorrected urine lowest, all adjustments by S-PFCRC and V-PFCRC exhibited levels between these two extremes. Furthermore, the minimum and maximum values for all CRN-adjusted modes showed moderate variation across both sexes and were not pushed into abnormal ranges by the various power-functional corrections. This confirms that both V-PFCRC and S-PFCRC do not introduce significant distortions compared to conventional methods, thereby maintaining reliable results.

Residual correlations between arsenic and CRN of complete data sets

Table 3 summarizes the Spearman correlations between urinary arsenic, age, and CRN. Sex-specific and combined data indicate a strong positive correlation between CRN and uncorrected arsenic levels. Conventional CRN correction (CCRC) reverses this to a negative correlation, indicating substantial overcorrection. In contrast, all power-functional correction modes—simple and variable—significantly reduce arsenic dependence on CRN. These results confirm the effective removal of dilution bias by all tested power-functional corrections, with no significant differences observed between simple and variable approaches.

However, as illustrated in Fig. 4a–c, reduced correlations in the overall dataset may mask distortions within specific exposure levels, as opposing biases can cancel each other out when aggregated.

Residual correlations between arsenic and CRN in exposure-stratified data

The efficacy of sex-differentiated V-PFCRC (mode 5) was validated using power-functional regression on linear data of septile-stratified arsenic subsets. Septile analysis was conductedfor uncorrected, CCRC, and V-PFCRC arsenic results (Fig. 3 and Supplementary Fig. 3). The near-parallel alignment to the x-axis and significantly lower slopes and R^2 values than uncorrected and CCRC results demonstrate that V-PFCRC efficiently removes bias in all exposure levels.

The Python/Jupyter code used for visualization is provided in *Excel File* 2, Sheet 8. Similar results were observed with log-transformed data (Fig. 4) for sex-specific and combined datasets using a lower CRN cutoff of ≤ 2.4 g/L. This analysis compared V-PFCRC with simple power-functional correction (S-PFCRC, mode 9) to highlight its advantages. Linear regressions on log-transformed data, as detailed in *Supplementary File* 3, accentuate and more effectively illustrate differences in curve characteristics. V-PFCRC outperforms other methods for both sexes and combined data, showing better alignment of regression lines with the x-axis and lower slopes and R^2 values. Uncorrected data have higher R^2 and slopes across all septiles, while CCRC overcorrects, producing significant inverse correlations with CRN. S-PFCRC, which uses a constant exponent (**b**), was less effective in septiles 1, 2, 6, and 7. In contrast, V-PFCRC effectively eliminated CRN-arsenic correlations across all septiles, offering superior dilution adjustments. While S-PFCRC performed reasonably well for the overall dataset (Table 3; *Excel File* 1, Sheet 5, F), V-PFCRC provided more consistent compensation for dilution errors, with the most notable improvements in the higher and lower ranges of the spectrum.

Comparison of sex-differentiated V-PFCRC with other result modes

The scatter and Bland-Altman plots in Supplementary Fig. 5 illustrate differences between V-PFCRC results (mode 5) and other adjustment modes across the dilution spectrum. The greatest scatter and standard deviations occurred between V-PFCRC and uncorrectedarsenic (uAsUC), particularly at mid-range CRN levels, with better alignment observed at the extremes. CCRC showed closer agreement with V-PFCRC than uAsUC but exhibited notable discrepancies at higher arsenic levels in females and lower levels in males. Other power-functional corrections, including S-PFCRC (mode 9) and sex-aggregated V-PFCRC (mode 6), were more closely aligned with sex-differentiated V-PFCRC. They displayed lower deviations on Bland-Altman plots and a characteristic bow-tie pattern. The downward tilt in females reflects a sex-specific relationship between exponent **b** and arsenic levels, driven by opposite signs of coefficient **c** at a CRN cutoff of ≤ 2.4 g/L. In summary, deviations from sex-differentiated V-PFCRC were most considerable for uncorrected urine (mid-range CRN), moderate for CCRC (higher CRN in females, lower CRN in males), and minimal for other power-functional adjustments.

Correlation between arsenic in EDTA blood and urine

Blood arsenic reflects combined exposure to organic and inorganic sources. Total arsenic in blood correlates well with urine levels, with CCRC further enhancing this correlation compared to uAsUC[55,61]. In Set 2, strong correlations were found between arsenic in EDTA blood and all urinary modes despite measurement uncertainty due to low concentrations. Of the 1,145 blood samples, 6% were below detection limits, and the 56.5% ≤ 1 µg/L had much lower resolution than urine values, as shown in Supplementary Fig. 7. Table 4 presents Spearman correlations between arsenic in blood and urine for sex-aggregated data, ranging from 0.81 (complete set) to 0.41 (top 10%). In line with previous studies, CCRC improved correlation coefficients by 0.1–0.14 compared to uAsUC[61]. While none of the six power-functional corrections outperformed CCRC in the complete dataset, several S-PFCRC and V-PFCRC modes showed better correlations in the top 50% and 25% subsets. Even in the top 10%, V-PFCRC modes 5 and 11 demonstrated significant improvements, with mode 5 consistently yielding the strongest correlations. Similar trends were observed in the sex-differentiate analyses in *Excel File* 1, Sheet 6, A and B.

Beyond correlation analysis, comparing sex ratios of quartile limits (1–3) between blood and urine provided additional insights into the dilution correction efficiency of the 14 result modes. Female-to-male ratios were calculated for blood and each urinary mode based on quartile limits determined for both sexes in Set 2 (Supplementary Fig. 8; *Excel File* 1, Sheet 5, G). Generally, females exhibited lower arsenic levels in both blood and urine compared to males, with the extent of this difference varying between the two matrices, as shown in Fig. 5. The most significant disparities across all correction methods (CCRC, P-PFCRC, and S-PFCRC) were found in the first quartile, indicating marked differences at lower arsenic levels. CCRC exhibited the largest discrepancy (+ 58%), likely due to its more pronounced overcorrection in higher diluted female urine samples (*Excel File* 1, Sheets 3 and 4, B). In contrast, simple (41–45%) and variable (37–47%) power-functional corrections showed more minor deviations in sex ratios from those observed in blood. At the median, sex ratios for blood and urine were closely aligned for all correction methods. In contrast, power-functional corrections outperformed uncorrected urine and CCRC for the third quartile, aligning better with blood values. Uncorrected urine (-22%) and CCRC (+ 23%) showed significant deviations from blood, while S-PFCRC and V-PFCRC demonstrated considerably more minor deviations (2–16%), with sex-differentiated and CRN-asymmetry compensated V-PFCRC modes 5 (+ 2%) and 11 (+ 3%) achieving the closest alignment for accurate dilution correction.

Correlation between arsenic in blood and age

Age was found to have a weak but significant positive correlation with blood arsenic in both sexes and sex-aggregated analyses, as shown in Supplementary Table 2. However, this correlation was only observed in the complete data but not in subsets where lower arsenic blood samples were excluded.

Correlation between blood arsenic and CRN

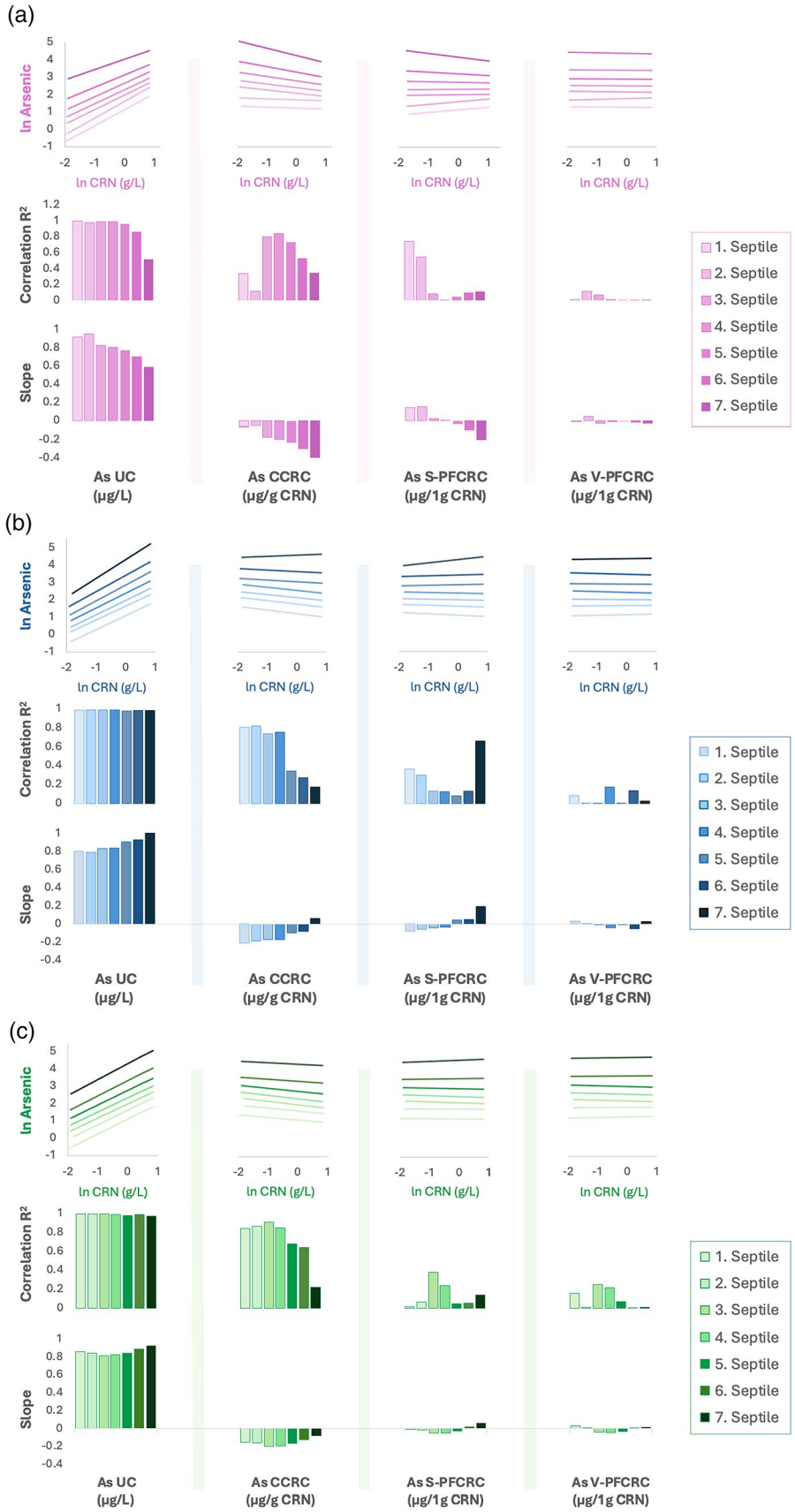
No significant correlation was found between urine CRN concentrations and blood arsenic in the analyzed sex-aggregated and male data sets (Supplementary Table 3). However, a weakly negative but significant correlation was observed for women in Set 2. This correlation was no longer observed when less precise values below the detection limit and lower blood arsenic samples were omitted.

V-PFCRC of other metals and iodine

The V-PFCRC method was validated for iodine and four metals (cesium, molybdenum, strontium, and zinc) across both sexes. As summarized in *Excel File* 2, Sheets 3–5, significant variations in curve fits (R^2) and coefficients were observed among the elements, indicating analyte-specific characteristics. Robust power-functional and log-linear relationships could be established for both sexes across all elements, supporting the method's generalizability. Effective SDAE removal by V-PFCRC is illustrated in Supplementary Fig. 2 for sex-aggregated, log-transformed data across all elements. With over 58,000 samples, iodine demonstrated the most robust curve fits (Supplementary Fig. 6), suggesting that larger datasets make SDAEs and V-PFCRC principles more apparent by averaging idiosyncrasies. The generally greater accuracy at lower CRN cutoffs (< 3.8 g/L) signifies more robust coefficients **c** and **d**, thus enhancing V-PFCRC formula accuracy.

Iodine analyses

Iodine was particularly suitable for validating the variable power functions correction method due to the availability of a large dataset—49,745 female and 8,694 male samples (*Excel File* 1, Sheets 7–9). This substantial sample size allowed for percentile ranking within narrower CRN and analyte bands, enabling a more detailed analysis of the relationship between iodine concentrations and CRN in highly concentrated urine.



Urinary iodine was correlated in uncorrected, conventionally, and four power-functionally CRN-corrected result modes. The two S-PFCRC or V-SPFCRC modes were tested in CRN-asymmetry-adjusted and unadjusted forms. Log-transformed, near-normally distributed CRN, urinary, and serum iodine data were analyzed.

◀**Fig. 4**. (**a**) Residual dependence of four urinary arsenic result modes on female CRN concentration. The top row depicts the linear regression curves between the log-transformed arsenic values over the log-transformed CRN values for the uncorrected (uAs_{UC}, mode 1), the conventionally CRN-corrected (CCRC, mode 2), the simple power functional CRN-corrected (S-PFCRC, mode 9), and the variable power functional corrected (V-PFCRC, mode 5) result mode for all respective seven septiles. The middle row illustrates the corresponding coefficients of determination (R^2). The bottom row shows the slope of the regression curve. Regression lines parallel to the abscissa, low R^2, and flat curve slopes indicate a weak residual dependence of the arsenic result on CRN and, thus, a more efficient dilution adjustment. (**b**) Residual dependence of four urinary arsenic result modes on CRN dilution in males. (**c**) Residual dependence of four urinary arsenic result modes on CRN dilution in sex-aggregate data.

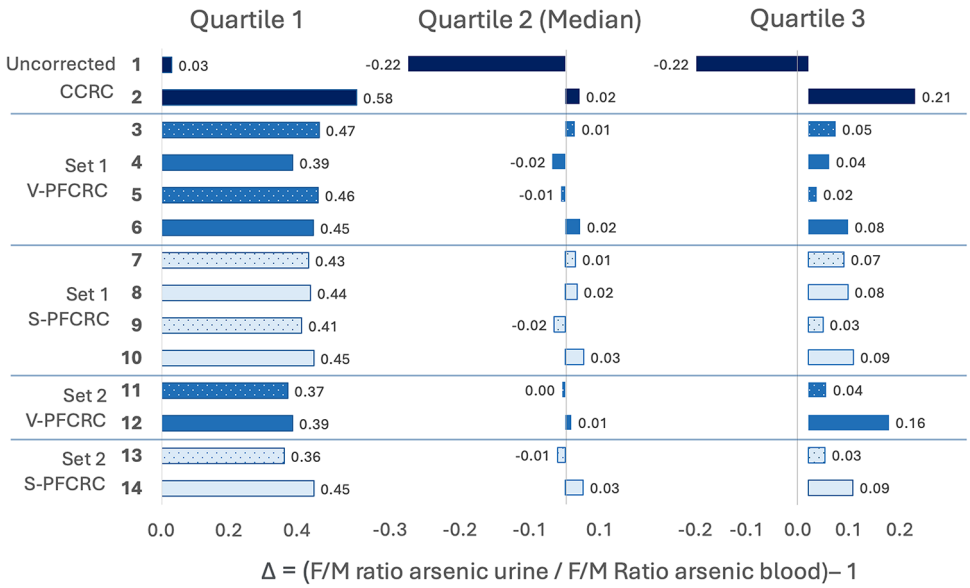


Fig. 5. Deviation of urinary from blood female/male ratios of arsenic quartile limits 1–3. The difference (Δ) between two F/M ratios can be expressed by the deviation of the quotient of both ratios from 1. This was calculated for all 14 urinary result modes as Δ = (FM ratio urine / FM ratio blood) − 1. Delta values above 0 indicate that the F/M ratio in the respective urine result mode is too high compared to the widely dilution-independent blood F/M ratios. A positive Δ can be caused by a dilution-adjustment-related upward distortion of female values, a downward distortion of male values, or any combination of distortions that shift the F/M ratio of the respective urine result mode net upward. Conversely, a Δ < 1 indicates the opposite.

Corresponding evaluations using non-parametric Spearman analyses and correlation of trimmed data sets are summarized in Supplementary Tables 4 and 5.

V-PFCRC of iodine

The regression curves for both steps of V-PFCRC demonstrated excellent fit across a wide range of CRN cutoffs, as shown in *Excel File* 2, Sheets 3 and 4 and illustrated for unrestricted CRN in Supplementary Fig. 6. Unlike arsenic, the log-linear slopes (exponent **b**) for iodine were consistently negative for both sexes, indicating decreased **b** with increasing iodine exposure. This trend suggests systemic dilution adjustment errors (SDAE) increase with higher urinary iodine levels, particularly in males, who exhibit a steeper decline (i.e., a more negative coefficient **c**). This contrasts with the more negative coefficient **c**, which is consistently observed in females for arsenic. These findings highlight distinct adjustment patterns between iodine and arsenic, emphasizing the importance of analyte- and sex-specific approaches in V-PFCRC analysis.

Table 5 summarizes Pearson correlations between 1,823 log-normal paired blood and urine iodine values across urinary result modes (*Excel File* 1, Sheet 9, B2). Both V-PFCRC methods improved correlations, particularly in females, likely due to their larger sample size ($n = 1{,}540$ vs. 283 males). Simple power function corrections performed similarly to CCRC in males but showed improvements in sex-aggregated and female subsets.

Correlation of serum and urine iodine across result modes

Similar to arsenic, improved correlations of iodine are also reflected in female-to-male (F/M) iodine quartile ratios for serum and urine, as illustrated in Fig. 6. Variable and simple power-functional CRN corrections substantially reduced F/M discrepancies in uncorrected and CCRC urine across quartiles. The more robust

Serum-Urine iodine Complete log data All serum iodine age 14–82 years	Sample Size	Pearson	Age	ln CRN (g/L)	ln iodine Uncorrected (µg/L)	ln iodine CCRC (µg/g CRN)	ln iodine V-PFCRC (µg/1 g CRN) CRN-asym. adjusted	ln iodine S-PFCRC (µg/1 g CRN) CRN-asym adjusted	ln iodine V-PFCRC (µg/1 g CRN) CRN-asym.	ln iodine S-PFCRC (µg/1 g CRN) CRN-asym.
Female + Male	1823	r	0.06	0.05	0.41	0.48	0.51	0.51	0.51	0.51
		p	0.008	0.04	< 0.001	< 0.001	< 0.001	< 0.001	< 0.001	< 0.001
Female	1540	r	0.03	0.07	0.39	0.44	0.47	0.48	0.47	0.47
		p	0.176	0.006	< 0.001	< 0.001	< 0.001	< 0.001	< 0.001	< 0.001
Male	283	r	0.16	0.02	0.51	0.61	0.62	0.61	0.62	0.61
		p	0.007	0.686	< 0.001	< 0.001	< 0.001	< 0.001	< 0.001	< 0.001

Table 5. Summary of Pearson correlations (r) and significance (p) of serum iodine with CRN and urinary iodine for complete female, male, and combined data sets.

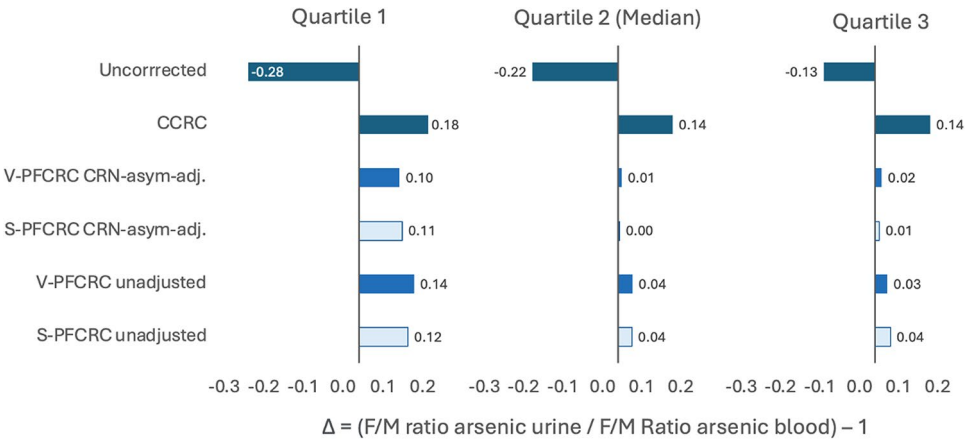


Fig. 6. Deviation of urinary iodine female/male ratios of arsenic quartile limits 1–3 from corresponding serum ratios. Analogy of the difference (Δ) between two F/M ratios to arsenic as shown in Fig. 4.

improvements in female urine-to-blood correlations for iodine and arsenic likely reflect more substantial SDAEs in more dilute female urine, with potential additional sex-specific factors warranting further investigation.

Computational assessment of arsenic V-PFCRC formula stability

Determinants of regression performance and curve stability

V-PFCRC accuracy depends on robust fits in the power-functional and log-linear steps. While R^2 is a suitable parameter for assessing curve stability, flat slopes significantly limit its reliability as a performance metric. In this study, power-functional slopes for arsenic ranged from 0.19 to 1.01, averaging 0.62 for females and 0.74 for males across CRN cutoffs of 2.4–3.8 g/L (*Excel File* 2, Sheets 2–3). In contrast, log-linear slopes were much lower (-0.27 to 0.09), averaging − 0.21 for females and − 0.03 for males, diminishing the utility of R^2 in these cases. Curve plots validating log-linear consistency beyond R^2 and slopes are provided in *Excel File* 2, Sheet 5. The following factors were found to influence R^2 and formula stability:

1. *CRN Cutoff Effects*: An in-depth Python/Jupyter analysis of the power-functional regression and log-linear relationship between normalized arsenic (= **a**) and exponents **b** across CRN cutoffs (≤ 2.4 to ≤ 4.4 g/L) is detailed in *Excel File* 2, Sheets 2–5. Curve fits in both regression steps, and the corresponding coefficients **c** and **d** of V-PFCRC formulas can vary by CRN cutoff. Higher CRN cutoffs reduced R^2 values in the power-functional step, especially in the highest exposure septile, but had no consistent impact on log-linear R^2.
2. *Outlier Filtering*: Stricter exclusion criteria (95th vs. 99th percentile) improved power-functional R^2, increasing averages from 0.74 to 0.86 in women and 0.82 to 0.89 in men. Log-linear R^2 showed minimal changes, except for strontium, where outliers distorted patterns (*Excel File* 2, Sheets 3 and 4).
3. *Coefficient Interdependencies*: Slope **c** is positively correlated (0.49) with power-functional R^2 and negatively correlated (− 0.53) with log-linear R^2. Converse correlations to those with **c** are observed for **d**, and the R^2 values of both regression steps, in concordance with a robust inverse association between **c** and **d** (-0.94).
4. *Sex Differences*: Log-linear R^2 was generally more robust in women, while power-functional R^2 was higher in men. Excluding the top 5% of analyte values resulted in average R^2 values of 0.86 (power-functional) and 0.56 (log-linear) for women and 0.89 (power-functional) and 0.52 (log-linear) for men.
5. *Sample Size and Consistency*: Larger datasets, such as those for arsenic and iodine, yielded the most consistent curve fits compared to other elements.

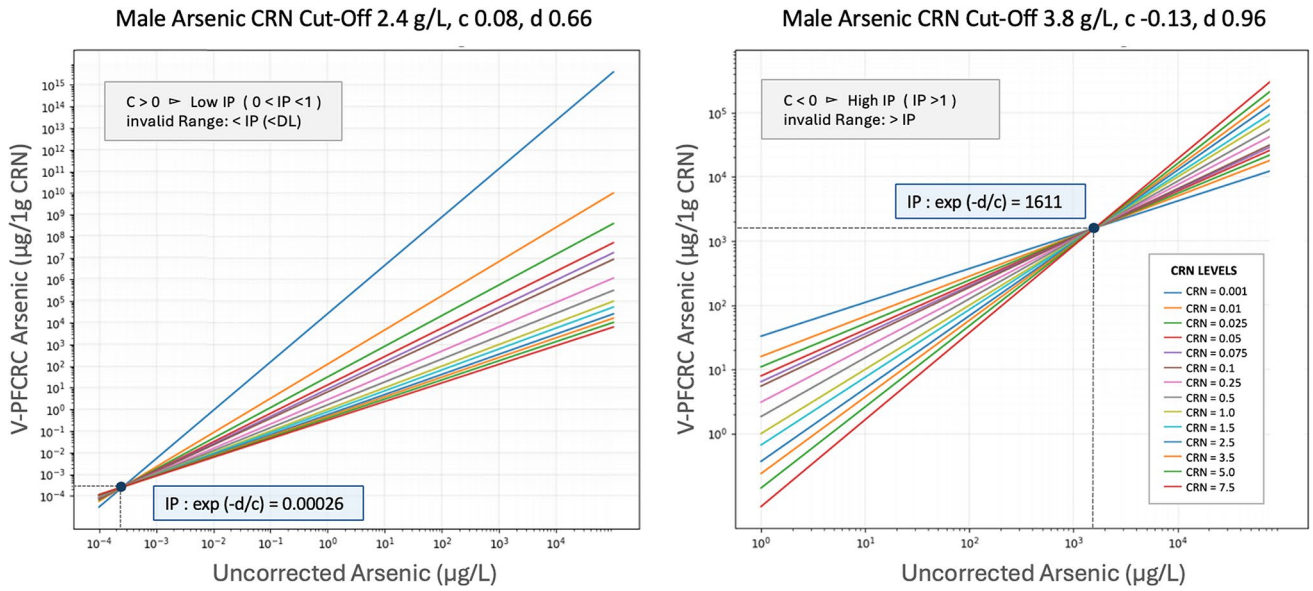


Fig. 7. Stability and Parameter Dependence of the V-PFCRC Correction. This figure illustrates the relationship between uncorrected analyte concentrations (A-UC) and V-PFCRC-adjusted values across a simulated CRN range (0.001–7.5 g/L), emphasizing parameter sensitivity. Coefficients **c** and **d** were derived from male urinary arsenic data using two CRN cutoffs: ≤2.4 g/L (positive **c**, left graph) and 3.8 g/L (negative **c**, right graph). All V-PFCRC adjustment curves for $\mathbf{c \neq 0}$ converge at an intersection Point (IP), calculated as $\mathbf{IP = exp\ (-d/c)}$. Positive coefficients **c** localize the IP between 0 and 1, below the detection limit (DL) for arsenic, preventing practically relevant curve crossings or paradoxical dilution adjustments. However, they may increase sensitivity to result fluctuations due to exponential divergence, particularly in highly dilute urine samples. Negative coefficients **c** shift the IP upward, restricting the valid analyte range to values between 1 and IP. While this range typically covers all or nearly all samples (> 99%), steep slopes (**c**) or high intercepts (**d**) can lower the IP and further narrow the valid range. Detailed crossing points for all elements' cutoff values and the Python code for their calculation are provided in *Excel File2*, Sheet 3.

These results demonstrate the influence of sample size, CRN cutoff selection, outlier filtering, and coefficient relationships on the stability and reliability of V-PFCRC formulas.

Influence of V-PFCRC formula variability on adjustment performance of V-PFCRC

Moderate coefficient variations, such as those seen between the V-PFCRC and S-PFCRC result modes tested (Supplementary Table 1) or between specific age groups[51], did not compromise the efficacy of dilution adjustments in entire collectives. Consistent improvements in bias removal (Table 3) and correlations with blood levels (Tables 4 and 5; Supplementary Fig. 8) were observed across all power-functional arsenic and iodine modes compared to uncorrected and CCRC urine data. The relationship between coefficients **c** and **d** in the V-PFCRC formula revealed three distinct adjustment patterns potentially stabilizing V-PFCRC:

1. *Strong relationships between coefficients **c** and **d**.*

The intercept **d** exhibited a robust inverse linear association with the slope **c** (r −0.93, $p < 0.001$), indicating inherent stabilization across diverse formula variants for a given biomarker (*Excel File* 2, Sheet 4, A-F, J).

2. *Element- and Sex-Specific Trends.*

Increasing the CRN cutoff generally decreases **c** while causing a compensatory rise in **d**. Arsenic and strontium in males showed unique patterns, as a shift to positive **c** values from predominantly negative **c** occurred when samples with CRN ≤ 2.6 g/L were excluded. This exclusion of highly concentrated urines results in steeper regression curves and disproportionately sharper increases in **b** values at elevated exposures (*Excel File* 2, Sheet 2), mathematically implying a corresponding rise in the coefficient **c**.

3. *Definition of Intersection Point and Valid Dilution Adjustment Analyte Ranges.*

The relationship between coefficients **c** and **d** of each V-PFCRC formula defines a unique intersection point (IP) where the uncorrected analyte concentration (in µg/L) equals the V-PFCRC analyte concentration in (µg/1 g CRN) (Fig. 7; *Excel File* 2, Sheets 3 and 9). The Intersection Point (IP), calculated as $\mathbf{IP = exp(-d/c)}$, is independent of CRN since all CRN curves of Fig. 7 converge at this point.

V-PFCRC behavior at extreme CRN and uncorrected arsenic values
The control calculations over a wide range of hypothetical arsenic and CRN concentrations in Fig. 7 demonstrated consistent positive associations between uncorrected analyte concentrations and V-PFCRC results across the spectrum. The functions remained stable, showing no unexpected directional changes, even at extreme CRN and arsenic levels. However, line crossings consistently occur at the intersection point (IP) for all V-PFCRC formulas with $\mathbf{c} \neq 0$. These crossings give rise to paradoxical corrections beyond the IP, with higher corrected values occurring at higher CRN levels and vice versa (*Excel File* 2 ,Sheets 3 and 9).

These effects manifest differently depending on the sign of **c**:

- **For c = 0**, no variation in coefficient **b** occurs, meaning no intersection of curves or IP exists in this case.
- **For c > 0**: paradoxical corrections appear below the IP (left graph in Fig. 7). Here, the IP is computationally restricted to values between 0 and 1, encompassing most samples within the valid range.
- **For c < 0**: paradoxical corrections occur above the IP (Fig. 7, right graph), with the IP position depending on the steepness of slope **c** and the size of **d**:
- In this case a steeper **c** (high absolute negative value) or more minor **d** results in a lower IP, narrowing the valid correction range, whereas flatter negative slopes **c** or higher **d** values shift the IP higher, broadening the valid range.

Despite the paradoxical and potentially misleading effects of dilution adjustments beyond the IP, these findings demonstrate that V-PFCRC formulas consistently maintain positive associations between uncorrected analyte concentrations and corrected results. All CRN curves steadily increase without local maxima or reversals, preserving a degree of interpretative validity even beyond the IP. The stability of CRN curves and the computational constraints of **c** and **d** provide a robust mathematical framework, while the paradoxical corrections beyond the IP underscore the importance of recognizing these limitations in extreme scenarios.

Impact of age-normalized CRN adjustments on V-PFCRC
Using CRN values adjusted to a 50-year age standard in V-PFCRC formula development preserved comparable curve fits across most ranges, with general improvements in females and declines in males, with moderately high R^2 values retained in most cases. Detailed curve data and coefficients for the 2.4–4.4 g/L CRN cutoff range are provided in *Excel File* 2 (Sheets 3–5). Age adjustments preserved the core principles of the V-PFCRC approach while modifying coefficients **c** and **d** as expected. The declining trend of **c** values with increasing CRN cutoffs observed in unadjusted CRN (Sheet 3) was maintained, with **c** variability increasing in females and decreasing in males. First-regression power-functional curve fits generally improved in females and slightly declined in males, with a similar trend observed in log-linear regression fits. Notably, age-adjusted CRN expanded the valid analyte range in most cases, as reflected by higher IP values across 11 tested CRN ranges in both sexes, except for one cutoff per sex, where IP was slightly lower.

Susceptibility of V-PFCRC and CCRC to CRN excretion variability
The analysis revealed substantial differences in how V-PFCRC and CCRC respond to fluctuations in CRN excretion (*Excel File* 2, Sheet 6). Using fixed urinary arsenic concentrations of 3, 10, and 40 µg/L, V-PFCRC significantly reduced deviations caused by CRN variability for both sexes, demonstrating greater stability than CCRC. This advantage became even more pronounced at higher uncorrected arsenic concentrations, as depicted in Supplementary Fig. 9.

V-PFCRC formula adjustments efficacy in addressing systemic CRN discrepancies
The effects of adjusted V-PFCRC formulas on mitigating systemic CRN excretion biases are detailed in *Excel File* 2, Sheet 7. Control calculations with original CRN values and ± 25% adjustments showed that systemic CRN fluctuations shifted arsenic values similarly for both CCRC and V-PFCRC across all seven exposure levels. While V-PFCRC reduced residual dilution errors within specific groups compared to CCRC (Figs. 3 and 4), it does not fully account for systemic biases between populations, such as females vs. males or carnivores vs. vegans. As discussed below, addressing these discrepancies may require further refinement using confounder-adjusted CRN values.

Discussion

Compensation of systemic dilution adjustment errors (SDAE) by V-PFCRC

Accurate biomonitoring depends on measurement data that reliably represent actual exposure levels. However, spot urine analysis often falls short of precisely compensating for hydration effects. Traditional methods like creatinine (CRN) normalization are prone to random errors driven by variability in CRN production and excretion (Table 1). Furthermore, nonlinear systemic dilution adjustment errors (SDAE) arise from the differing responses of biomarkers and standard correctors to diuresis, limiting the effectiveness of adjustments based on CRN, osmolality, or specific gravity. Even advanced multilinear covariate models are influenced by non-linear SDAEs, highlighting the complexity of achieving accurate normalization[3].

The Variable Power-Functional Creatinine Correction (V-PFCRC) method introduced here addresses these challenges by recalibrating the relationship between CRN and analytes across varying dilution conditions. Using CRN-specific percentiles and standardizing analyte values to a CRN level of 1 g/L, V-PFCRC effectively neutralizes nonlinear biases and ensures more accurate corrections across diverse exposure and hydration scenarios. This innovative approach improves consistency and reliability in biomonitoring by resolving the complexities of SDAEs. V-PFCRC's efficacy was demonstrated on 5,553 arsenic samples, significantly reducing CRN-related biases and improving urine-to-blood correlations. Validation across additional datasets—

including cesium, molybdenum, strontium, zinc, and iodine—further confirmed its ability to minimize CRN-related biases, particularly enhancing serum-urine correlations for iodine. Unlike streamlined methods such as S-PFCRC, which rely on fixed exponents, V-PFCRC also delivers robust adjustments at analyte extremes (Figs. 3 and 4). Despite moderate coefficients **c** and **d** variations by sex, age, or methodological adjustments, V-PFCRC consistently outperformed traditional methods, providing more precise dilution adjustments[3,51].

The advantages of V-PFCRC are most apparent in inter- and intra-individual comparisons. For instance, as shown in Supplementary Fig. 10, moderate arsenic exposure could cause uncorrected levels to vary by a factor of 5.9 within the accepted CRN range of 0.3 to 3 g/L. In this example, the traditional CRN correction reduces this variability to a factor of 1.7, but biases persist due to considerable overcorrection. In contrast, V-PFCRC consistently normalizes exposure levels to the expected value of 1 g/L CRN, ensuring stability across the entire dilution range. Although dilution errors in uncorrected or CCRC urine tend to average out in more extensive cohort studies, enhancing normalization accuracy by V-PFCRC reduces sample size requirements and strengthens statistical power.

This study's large sample size enabled a comprehensive evaluation of how exposure levels, age, and sex influence SDAE of total weight arsenic (TWuAs) and its compensation through V-PFCRC[51]. By accounting for combined SDAEs across all TWuAs compounds and their proportional variability, V-PFCRC formulas demonstrated robustness in arsenic normalization. The results of this study align with the nonlinear arsenic correction findings of Middleton et al. (2019), derived from a regionally distinct population where arsenobetaine was excluded, in contrast to its inclusion here[3]. The consistency of S-PFCRC exponents (0.75–0.89) with the previously reported value of 0.8 further supports the validity of these findings. The slightly higher average exponent (0.83) observed in this study may be attributed to the rapid excretion of arsenobetaine, which is less efficiently reabsorbed and thus behaves more similarly to CRN than inorganic arsenic and its principal metabolites.

Population-specific factors—such as variations in diuresis, CRN excretion (Table 1), biomarker composition, and genetic or epigenetic influences on urinary analyte toxicokinetics—may occasionally require formula adjustments for greater precision. However, these modifications are typically minor and unlikely to affect group-level results substantially. The Python codes provided in Excel File 2 (Sheets 3–5 and 8) facilitate efficient assessments and refinements of V-PFCRC formulas for specific populations, underpinned by a sufficiently robust and representative dataset. As more extensive sample sizes are analyzed, machine learning mechanisms could further accelerate the evolution of V-PFCRC by identifying patterns and optimizing coefficients **c** and **d**. This will enable continuous refinement and improved accuracy in normalization.

Determinants of the analyte-specific coefficients c and d

The analyte-specific coefficients **c** and **d** are critical for achieving accurate, analyte-specific V-PFCRC corrections across various analytes and populations. These coefficients are shaped by the interplay between tubular dynamics, CRN distribution, and analyte composition within the reference collective. Understanding these determinants is essential for optimizing the accuracy and applicability of V-PFCRC formulas to diverse datasets, thereby enhancing correction precision across different study populations.

1. Tubular Excretion Behavior

   Renal handling of analytes significantly influences their excretion. At elevated analyte levels in the peritubular capillaries, tubular transport may become saturated, reducing net secretion relative to creatinine (CRN) and resulting in negative **c** values. Cesium, iodine, molybdenum, and zinc consistently show negative **c** values across all CRN cutoffs in both sexes. Arsenic and strontium, however, occasionally exhibit positive c values, particularly in male samples when using lower CRN cutoffs, though their overall trend remains negative. Negative c-values may reflect diminished tubular excretion efficiency relative to creatinine (CRN), particularly under conditions of poor hydration and higher exposure levels. In such cases, the reduced glomerular filtration (GFR) and urinary flow rate (UFR) shift excretion reliance toward tubular mechanisms (Fig. 1a).

2. Urinary CRN in study collectives and V-PFCRC formulas

   The average concentration and distribution of CRN within a dataset significantly influence the coefficients **c** and **d**. Higher CRN levels can saturate tubular excretion[44], shifting the balance relative to analytes and potentially contributing to sex-specific differences in V-PFCRC formulas. However, variations in CRN distribution alone do not fully account for the observed differences in **c** and **d**. Even after correcting for CRN asymmetry, sex-specific variations in arsenic excretion remain, likely due partly to enhanced arsenic methylation in females driven by estrogen[49,51,62,63]. This highlights the critical need to select an appropriate reference dataset to optimize the accuracy of V-PFCRC corrections.

3. Analyte composition and concentration

   As mentioned, analyte chemical composition can vary with concentration, affecting renal excretion patterns across different exposure levels. For example, higher concentrations of poorly absorbable arsenobetaine in urine can increase the exponent b, resulting in a more positive c value. This underscores how shifts in analyte composition can impact renal handling, with significant implications for V-PFCRC corrections.

Limitations in V-PFCRC

Through its dual regression process, the V-PFCRC method significantly reduces systemic dilution adjustment errors (SDAE), eliminating the need for strict creatinine (CRN) cutoffs (0.3–3 g/L) required for conventional creatinine correction (CCRC)[48]. However, like all correction methods, V-PFCRC may encounter limitations, particularly at extreme CRN or analyte levels. These limitations are explored in detail below.

CRN range considerations in V-PFCRC analysis

While polynomial adjustments theoretically risk producing erratic behavior beyond the established range, Python-based control calculations (*Excel File* 2, Sheet 9) confirm steady increases in corrected analyte concentrations as uncorrected analyte levels rise. These calculations, illustrated in Fig. 7, demonstrate that V-PFCRC maintains a stable and predictable response without local maxima or slope changes, even across extreme ranges of CRN (0.001–7.5 g/L) or uncorrected values (0–10,000 µg/L), which are rarely encountered in real-world scenarios. At the lower extreme of the CRN range, caution is warranted for highly dilute samples (CRN < 0.025 g/L). The markedly steeper slope of the CRN 0.001 g/L curve (left graph, Fig. 7) highlights the potential for significant discrepancies arising from small measurement fluctuations. These discrepancies will increase with analyte concentrations, especially when positive coefficients **c** lead to exponential divergence of CRN curves. To ensure optimal precision, highly diluted urine samples with CRN < 0.05 g/L should be excluded from analysis, and repeat testing is recommended.

In an extensive iodine dataset comprising 59,791 samples, only one sample (CRN = 0.01 g/L) fell below this threshold (Excel File 1, Sheets 16 and 17), illustrating the rarity of such cases. At the upper extreme, highly concentrated urine (CRN > 3.4 g/L in females and CRN > 3.8 g/L in males) may exhibit moderate downward shifts in V-PFCRC results due to increasing deviations of actual data distributions from power-functional regression curves beyond these limits (Supplementary File 3). These shifts are likely caused by tubular net excretion saturation resulting from reduced urinary flow, which prolongs tubular retention and increases analyte reabsorption. This process shifts analyte/CRN ratios toward CRN, maintaining a constant excretion rate even at slower urinary flow (Fig. 1).

Adopting the recommended V-PFCRC cutoffs to the iodine dataset (*Excel File* 1, Sheets 16 and 17) substantially reduced sample rejection rates—from 22.2 to 0.89% in females (a 24-fold reduction) and from 10.9 to 1.31% in males (an 8.3-fold reduction).

Figure 3 and Supplementary Fig. 1 illustrate the advantages of V-PFCRC over CCRC and uncorrected values when these CRN cutoffs are applied in formula generation.

Analyte range considerations in V-PFCRC analysis

The Intersection Point (IP) in V-PFCRC formulas, defined as $\mathbf{IP = exp(-d/c)}$, marks the boundary where uncorrected and V-PFCRC-adjusted analyte concentrations converge across all CRN levels (Fig. 7). Beyond this point, paradoxical adjustments may occur, with higher CRN values artificially elevating corrected results and lower CRN values suppressing them. These artifacts not only fail to mitigate dilution bias but may also amplify it, while narrowing the valid analyte range. Steeper negative **c** values or smaller **d** values further decrease the IP, thereby increasing susceptibility to paradoxical effects.

Despite moderate coefficient variability, most equations position the IP to cover over 99% of samples, demonstrating practical applicability (*Excel File* 2, Sheet 3). Aligning uncorrected analyte ranges with the IP improves correction accuracy and minimizes bias. Extreme outliers remain readily identifiable as excessively high across all result modes, including uncorrected values and paradoxically adjusted V-PFCRC beyond the IP, due to the positive relationship between uncorrected and V-PFCRC results. For optimal precision, values exceeding the IP when applying negative **c** should either remain uncorrected or be interpreted using CCRC or S-PFCRC. While less accurate than V-PFCRC for most samples, these methods avoid paradoxical corrections and may handle extreme outliers more effectively, depending on deviations from typical ranges and the magnitude of systemic dilution errors (SDAEs) in traditional methods. Automating outlier flagging above the intersection point (IP) and applying alternative corrections could further enhance the utility of V-PFCRC in routine practice analyses.

Inclusion criteria and curve fitting in V-PFCRC formula determination

Good alignment of all curves with the data in both regression steps is essential for effective V-PFCRC biomarker correction. While high R^2 values in power-functional curves typically indicate robust curve fits, R^2 alone in the second log-linear regression step is a less reliable predictor of robustness. As discussed below, R^2 should be assessed alongside coefficients **c** and **d** and validation plots (*Excel File* 2, Sheet 5). Achieving optimal fits requires careful consideration of CRN cutoff selection, outlier filtering, coefficient dependencies, and inherent data variability.

Impact of CRN cutoffs on V-PFCRC formulas

High creatinine (CRN) cutoffs are significantly negatively correlated with R^2 values of power-functional curves (r -0.42, $p < 0.001$) of the first regression step, particularly at higher exposure levels in the seventh septile (*Excel File* 2, Sheets 2 and 4, J). This likely reflects partial saturation of tubular net secretion of analytes at elevated levels and reduced glomerular filtration rates (GFR) in concentrated urine, leading to flattened data distributions (Supplementary File 3). In contrast to the first step, CRN cutoffs have no significant influence on the log-linear second step (*Excel File* 2, Sheet 4, J), where small slopes (**c**) make R^2 a less reliable measure of curve alignment with data points, despite good overall curve fitting.

Constraining CRN ranges significantly influences V-PFCRC formulas by affecting curve fits, coefficients, and intersection points (IPs), as summarized in *Excel File* 2, Sheets 2–5. These differences appear to be

confounded by age effects on CRN, as adjusting CRN to age 50 reduces variability in coefficients **c** and **d** across CRN cutoffs (*Excel File* 2, Sheet 10). A strong inverse linear relationship between coefficients **c** and **d** reduces extreme formula fluctuations. Higher CRN cutoffs decrease **c** and increase **d**, lower the IP, and narrow the valid dilution adjustment range. Despite these variations in formulas, the neutralization of CRN bias and blood-urine correlations remains robust at the group level, with V-PFCRC outperforming uncorrected and conventional creatinine ratio correction (CCRC) methods (Table 3; Supplementary Fig. 5). The Python tools provided in Excel File 2 enable rapid assessment of V-PFCRC applicability and optimal CRN cutoff determination for any spot urinary biomarker, streamlining analysis and minimizing manual redundancies.

Outlier filtering

Stricter exclusion criteria (e.g., 95th vs. 99th percentile) significantly enhance power-functional curve fits, increasing the average R^2 from 0.76 to 0.87 in women and 0.85 to 0.90 in men (*Excel File* 2, Sheet 4, H). However, these criteria have a limited impact on log-linear regression fits. Strontium is a notable exception, where asymmetrically distributed outliers obscure V-PFCRC regularity, requiring stricter filtering (*Excel File* 2, Sheets 3 and 4, I). Similarly, the female zinc formula calculated with a CRN cutoff of ≤ 2.4 g/L was skewed by outliers, resulting in a steep slope **c** (− 1.34), a high intercept **d** (8.68), and a moderate R^2 (0.62). Increasing CRN cutoffs and/or applying stricter filtering (> 95th percentile) resolved this issue (*Excel File* 2, Sheet 4, G–I).

Coefficient interdependencies

In both sexes and combined data, R^2 values of power-functional regressions are significantly ($p < 0.001$) positively correlated with slope **c** (combined: 0.6, female: 0.73, male: 0.39) and negatively with intercept **d** (*Excel File* 2, Sheet 4, J). Conversely, in log-linear regressions, R^2 values correlate negatively with **c** and positively with **d**. A higher **c** is associated with better first-step curve fits and broader analyte ranges, but it results in lower log-linear R^2 values. This reduction is primarily due to predominantly negative **c** values approaching zero, which does not indicate poorer curve fits, as mentioned.

Sex-based differences

Due to more highly concentrated urine, men generally exhibit higher power-functional regression stability, while log-linear regressions are more robust in women. With the top 5% of analyte values excluded, R^2 values for women average 0.87 (power-functional) vs. 0.54 (log-linear) and 0.90 vs. 0.53 for men (*Excel File* 2, Sheet 4, H).

Challenges with low variability and erratic distributions

Poor curve fits and low R^2 values in the second regression step can arise from three primary factors:

1. **Limited Sample Size**: Smaller datasets produce less defined patterns, allowing randomness to dominate and resulting in weaker curve fits.
2. **Inherently Low Variability**: Minimal variation in the exponent **b** across exposure levels results in low R^2 despite well-aligned curves. For example, when **c** is close to zero (e.g., male arsenic data), curves may align well but yield low R^2 because **b** does not vary across exposure levels (*Excel File* 2, Sheet 5). In such cases, forcing a log-linear fit is unnecessary and offers no advantage over using a fixed exponent in a simplified S-PFCRC model.
3. **Inherently Erratic Behavior**: Erratic distributions may stem from nonlinear biases, variations in chemical forms, or differing exposure sources.

In the latter case (3), systemic dilution correction errors will persist depending on exposure level. Forcing log-linear fits in cases of erratic behavior, albeit with sufficient sample sizes (Case 3), can risk inaccurate adjustments. Instead, a fixed exponent **b** might be used in a S-PFCRC model:

$$\text{Standardized value at 1g/L CRN} = \text{Analyte Uncorrected/CRN}^{\,b}$$

This approach ensures robust corrections when data lack reliable log-linear relationships, outperforming the flawed fixed **b** = 1 exponent in conventional creatinine ratio correction (CCRC). The optimal S-PFCRC coefficient **b** can be computed from previous methods[3], or estimated by averaging **b**-values from stratified power functional curves.

Concluding remarks on addressing V-PFCRC limitations

Understanding and carefully balancing the partially competing limitations outlined above is necessary to determine the optimal V-PFCRC formula for any biomarker. This ensures that the V-PFCRC method remains robust and adaptable across diverse biomonitoring applications. Refining CRN cutoffs, improving outlier management, and aligning analyte ranges with Intersection Points enhances correction accuracy while minimizing bias. These refinements provide critical insights for advancing future applications of V-PFCRC in clinical and epidemiological research, where precision and data retention are paramount. The Python-based tools provided in *Excel File* 2 further reinforce the method's utility by enabling rapid formula validation and ensuring reproducibility in large-scale studies.

Potential for V-PFCRC formula refinement

While V-PFCRC substantially enhances the accuracy and reliability of traditional methods, certain aspects may benefit from additional optimizations, depending on the research context. Potential enhancements include:

V-PFCRC formula adjustment to systemic group differences

Systemic biases in CRN excretion complicate comparisons among groups with inherently different CRN production levels, such as females versus males or carnivores versus vegans. While adjusting V-PFCRC formulas effectively reduces systemic dilution adjustment errors (SDAE) within subgroups, it is less effective at correcting disparities among groups with distinct baseline CRN levels. Even after formula adjustments, populations with lower CRN Excretion still exhibit higher mean CCRC and V-PFCRC values than those with higher CRN excretion levels, as illustrated in *Excel File 2: V-PFCRC Formula Validation*, Sheet 6. The CRN adjustments detailed in the following section can further mitigate these remaining group biases.

Adjusting V-PFCRC to CRN production and excretion confounders

Multiple factors, including muscle mass, age, gender, and diet, influence creatinine (CRN) production and excretion. While V-PFCRC corrections effectively reduce SDAEs and other nonlinear errors, they do not primarily address biases arising from individual variability in CRN production and excretion. Despite this, control calculations (Supplementary Fig. 9; *Excel File* 2, Sheet 6) demonstrate that V-PFCRC shows greater resilience than CCRC against CRN excretion biases compounded by these factors (summarized in Table 1).

The complex interplay of these confounders tends to moderate their overall impact, and prior research suggests that nonlinear SDAEs—directly addressed by V-PFCRC—may contribute more significantly to dilution adjustment limitations than cumulative variability in CRN excretion[3]. This is evidenced by the convergence of osmolality and specific gravity benefits over CRN when using S-PFCRC instead of CCRC[3].

Despite its substantial accuracy improvements, V-PFCRC remains affected by residual distortions from CRN confounders, particularly at low exposure levels (*Excel File* 2, Sheet 6). These distortions underscore the need for further refinement of the V-PFCRC approach, as illustrated in the next section by the example of CRN age bias.

Addressing CRN confounders via V-PFCRC – a case study on age

By adjusting CRN to its critical confounder, age, this approach not only preserved but also enhanced the stability and accuracy of the V-PFCRC method, demonstrating its adaptability and potential for further refinements. This effect is particularly evident in stabilizing V-PFCRC curves and reducing coefficient variability across a wide range of CRN cutoffs when using CRN adjusted to an age of 50 years instead of the original CRN for formula calculations (*Excel File* 2, Sheet 10). Age-related effects appear to confound coefficient fluctuations as higher cutoffs exclude more high-concentration samples, disproportionately affecting younger populations with naturally higher CRN levels[51]. This introduces an age bias, resulting in curves at lower CRN cutoffs more reflective of older populations. Therefore, the proposed age-adjustment approach can mitigate this variability and strengthen formula robustness while ensuring broader applicability.

While age was used here as a case study due to data availability, this approach is readily extendable to other confounders, such as muscle mass, diet, and physical activity, which significantly influence creatinine variability. These refinements, discussed below, may pave the way for even greater precision and versatility in biomonitoring applications.

Addressing combinations of confounders in CRN production

Building on the potential for single-confounder adjustments, the methodology presented below outlines a strategy for accounting for combinations of confounders to further enhance V-PFCRC accuracy.

Step 1: Calculate Adjustment Factors.

Three key factors are considered here to adjust for individual variability in CRN excretion:

1. Age Factor (F_{age}): Adjusts for changes in CRN levels due to age.
2. Muscle Mass Factor (F_{muscle}): Accounts for differences in muscle mass that influence baseline CRN levels.
3. Dietary Factor (F_{diet}): Corrects for dietary effects on CRN excretion, such as meat consumption versus plant-based diets.

Example of Dietary Adjustment.

If a diet high in protein (e.g., meat-heavy diet) increases CRN by 20% above the baseline, the dietary factor would be set to 0.8 (i.e., 1–0.2) to adjust for this effect:

$$F\ diet = Standard\ Diet/Individual\ Diet$$

Step 2: Adjusting the Observed CRN Value.

In case of independence and linearity, the adjusted CRN could be calculated by multiplying various factors:

$$CRN_{adjusted} = CRN \times F_{age} \times F_{muscle} \times F_{diet}$$

Step 3: Substituting CRN with $CRN_{adjusted}$ in the V-PFCRC Formula.

This adjusted CRN aims to account for individual variability in age, muscle mass, and diet. However, there are crucial considerations to address here:

1. **Adjusting for multiple factors assumes independence, which is invalid in this context.** For instance, the interrelated factors reduced muscle mass, lower protein intake, and decreased physical activity in older individuals collectively lower urinary CRN levels, compounded by a decline in renal concentration ability. Adjusting for such correlated variables can introduce collinearity issues, potentially leading to distortions and biased estimates[38]. Thus, focusing on a single factor or using a combined metric may provide a more accurate and reliable model.
2. **Accurately determining lean muscle mass remains challenging.** The body mass index (BMI) has been used as a proxy for muscle mass in covariate models adjusting for dilution[10,49,64]. However, BMI does not reliably differentiate between muscle and fat, limiting its precision[65]. More sophisticated methods like dual-energy

X-ray absorptiometry (DEXA), bioelectrical impedance analysis (BIA), or BMI-adjusted calf circumference offer greater accuracy. Still, they are often cost-prohibitive and impractical for routine clinical or epidemiological use[66]. Additionally, factors like hydration status, age, sex, or population-specific characteristics can affect these measurements, potentially introducing significant errors if unadjusted[67]. Moreover, estimating dietary intake is complex and requires extensive pre-analytical surveys, adding effort and uncertainty.

Given these intricacies, adjusting V-PFCRC formulas for factors beyond age is unlikely to yield substantial gains in accuracy relative to the additional effort and sophistication required. In most clinical and epidemiological contexts, a straightforward age adjustment may strike the optimal balance between accuracy and practicality, preserving V-PFCRC's broader efficacy in managing dilution-related biases.

Conclusion

Hydration-dependent fluctuations present significant challenges for conventional dilution adjustments in spot urinary biomarker analysis, as existing methods cannot fully account for these variations. This study introduces the V-PFCRC method, effectively reducing residual dependence on creatinine for all six spot-urinary elements tested and markedly enhancing urine-to-blood correlations for arsenic and iodine. By addressing biases across varying concentration ranges, V-PFCRC surpasses both conventional (CCRC) and simplified power-functional approaches (S-PFCRC), establishing a more reliable framework for managing dilution-related biases. This novel method offers a robust numerical solution to mitigate residual dilution errors, reducing sample rejections and significantly improving the accuracy of spot urinary assessments across a broad range of CRN levels. Furthermore, V-PFCRC is adaptable to other correction methods, such as osmolality and specific gravity. V-PFCRC formulas can be efficiently established using Python/Jupyter platforms and seamlessly integrated into existing laboratory software without straining infrastructure. This integration enhances precision and interpretative power, positioning V-PFCRC as a valuable tool in clinical, epidemiological, and forensic research.

Data availability

Excel File 1: Data Sets contains all datasets used and analyzed during this study. Additional details can be obtained from the corresponding author upon reasonable request.

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Author contributions
The author was responsible for designing the study, collecting data, reviewing the literature, conducting statistical analysis, developing corrective formulas, and writing the manuscript.

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Declarations

Competing interests
The authors declare no competing interests.

Ethics approval and consent to participate
The Ethics Committee of Northwest and Central Switzerland (EKNZ), Hebelstrasse 53, 4056 Basel, Email: eknz@bs.ch, deemed the study exempt from ethical approval and the need for informed consent. This was because the study involved a retrospective error analysis of anonymized data, including urinary creatinine, total weight arsenic, cesium, iodine, molybdenum, strontium, and zinc, EDTA blood arsenic, serum iodine, age, sex, and exam date, without any further clinical information. The purpose was methodological quality improvement only. No pharmacological or other medical intervention, no risk for physical or psychological harm, identifiability, or breach of confidentiality for any subject involved. The experiments followed the guidelines and regulations provided in the Declaration of Helsinki.

Consent for publication
Not applicable.

Additional information

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